



UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH

Escola Superior d'Enginyeries Industrial,
Aeroespacial i Audiovisual de Terrassa

Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model

Document:

Drawings

Author:

Sabina Campos i Fuentes

Director/Co-director:

Oriol Lordan Gonzalez

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BACHELOR FINAL THESIS

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List of acronyms

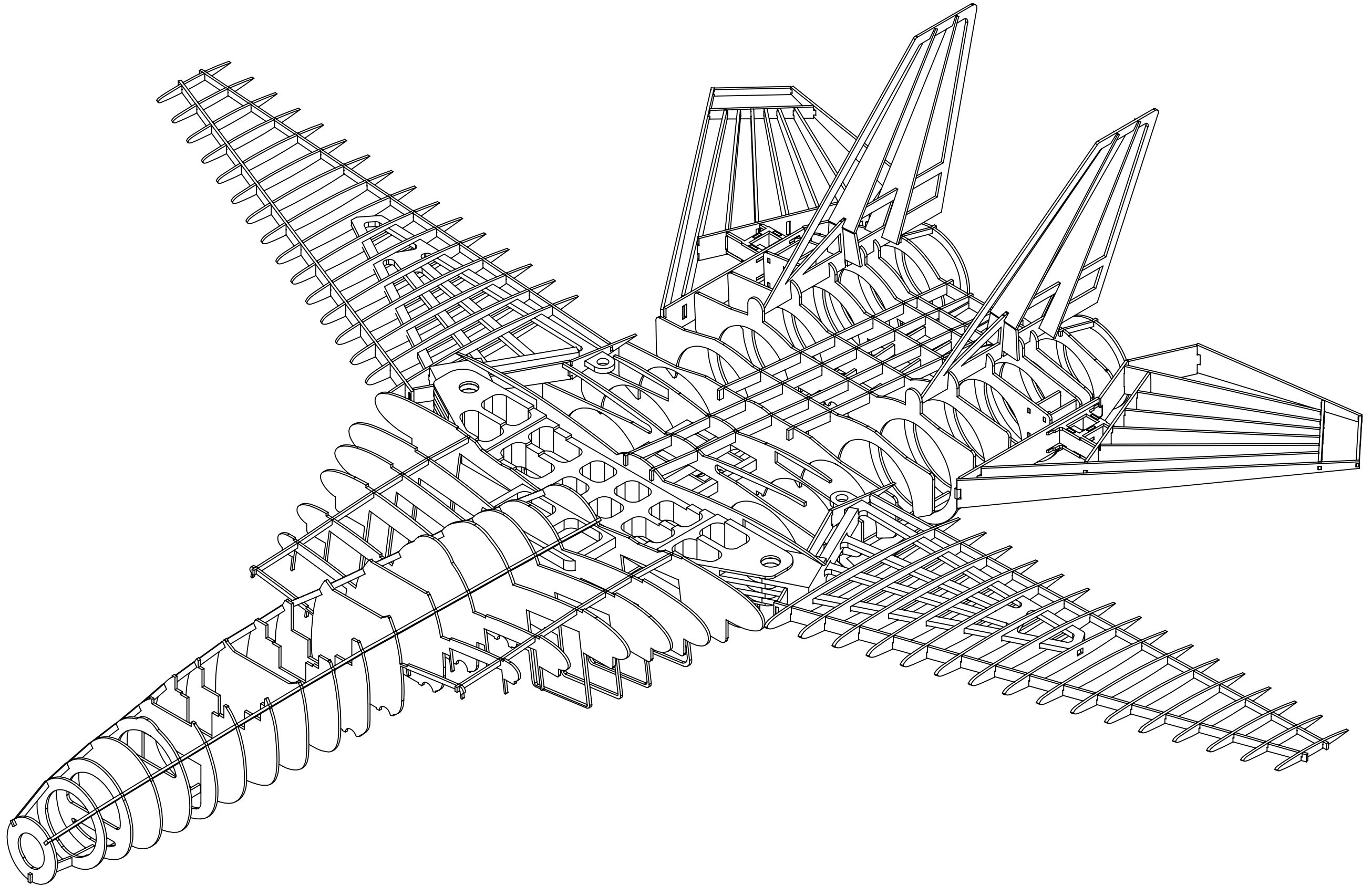
CS	Cross-section
B	Beam
R	Rib
WB	Wing beam
WC	Wing connector
WS	Wing sweep
PM	Pivot mechanism
VS	Vertical stabilizer
VF	Ventral fins
HS	Horizontal stabilizer

1 Introduction

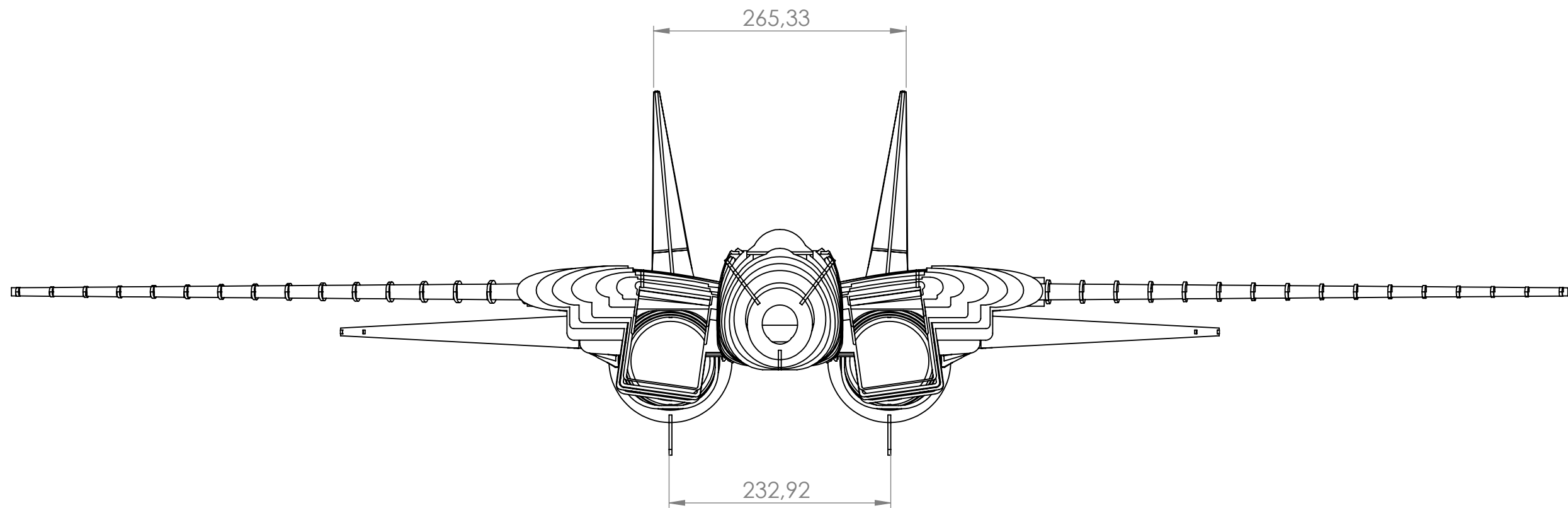
This document presents the design drawings of the Grumman F-14 Tomcat scale model that has been carried out during the completion of the thesis. These have been extracted from the 3D model made in SolidWorks. All the dimensions shown are in millimeters. In the box of each sheet the scale of the parts is specified, as well as the sheet number. In the drawings only the front view has been dimensioned since the thickness of all the pieces is 3mm.

The first four sheets correspond to the general plans of the set, representing the isometric view and the three main views. Then you can find the drawings of each structural element separately. It should be mentioned that there are some curves that could not be dimensioned because they are splines.

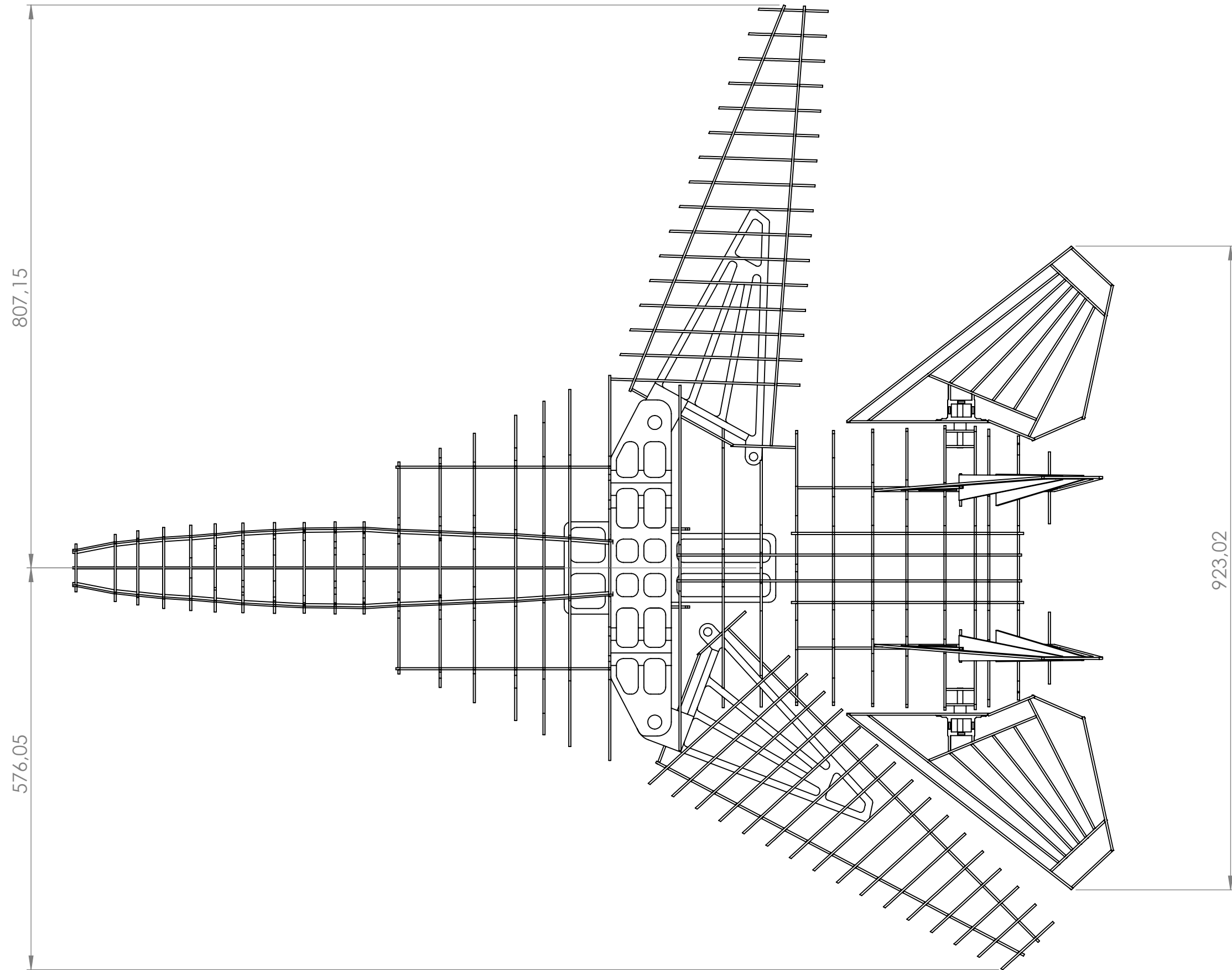
At the end, the drawings obtained with AutoCAD of all the parts with the necessary color layers for laser cutting are exposed. The original files are 1:1 scale in millimeters.



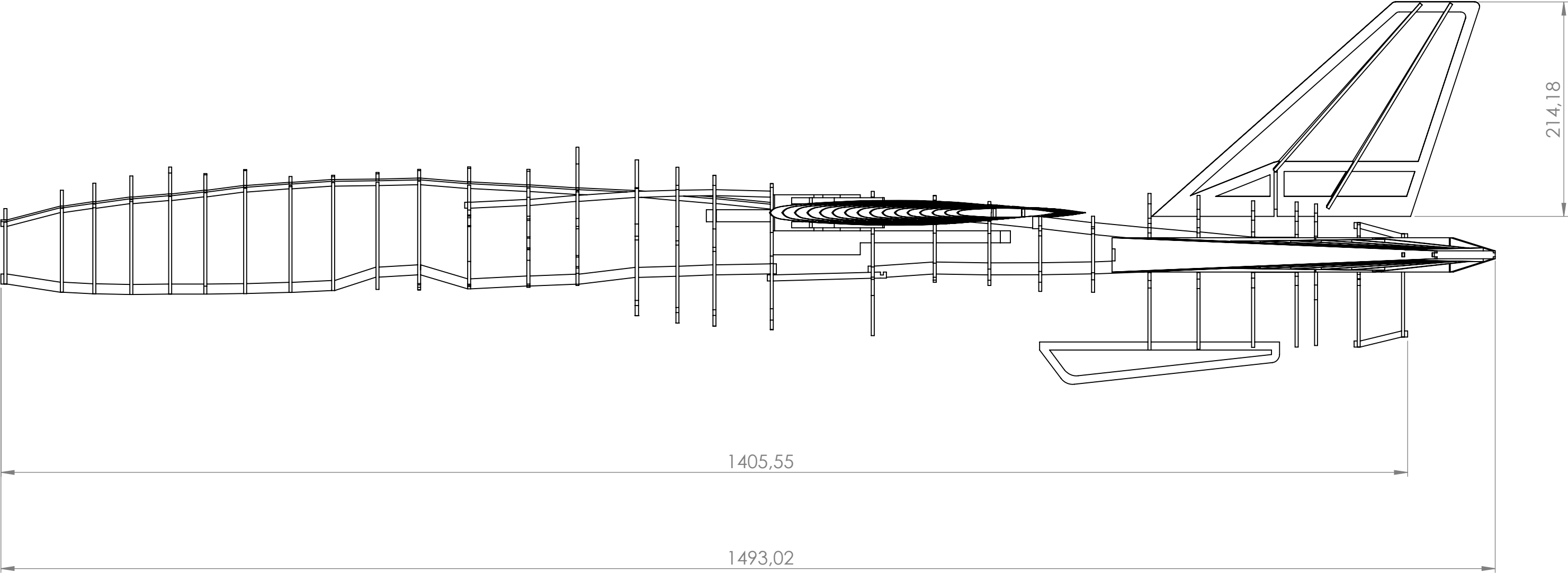
TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:4	Date: 21/06/2023	Size: A3
Sheet: 1	Name: Sabina Campos i Fuentes	UPC - ESEIAAT



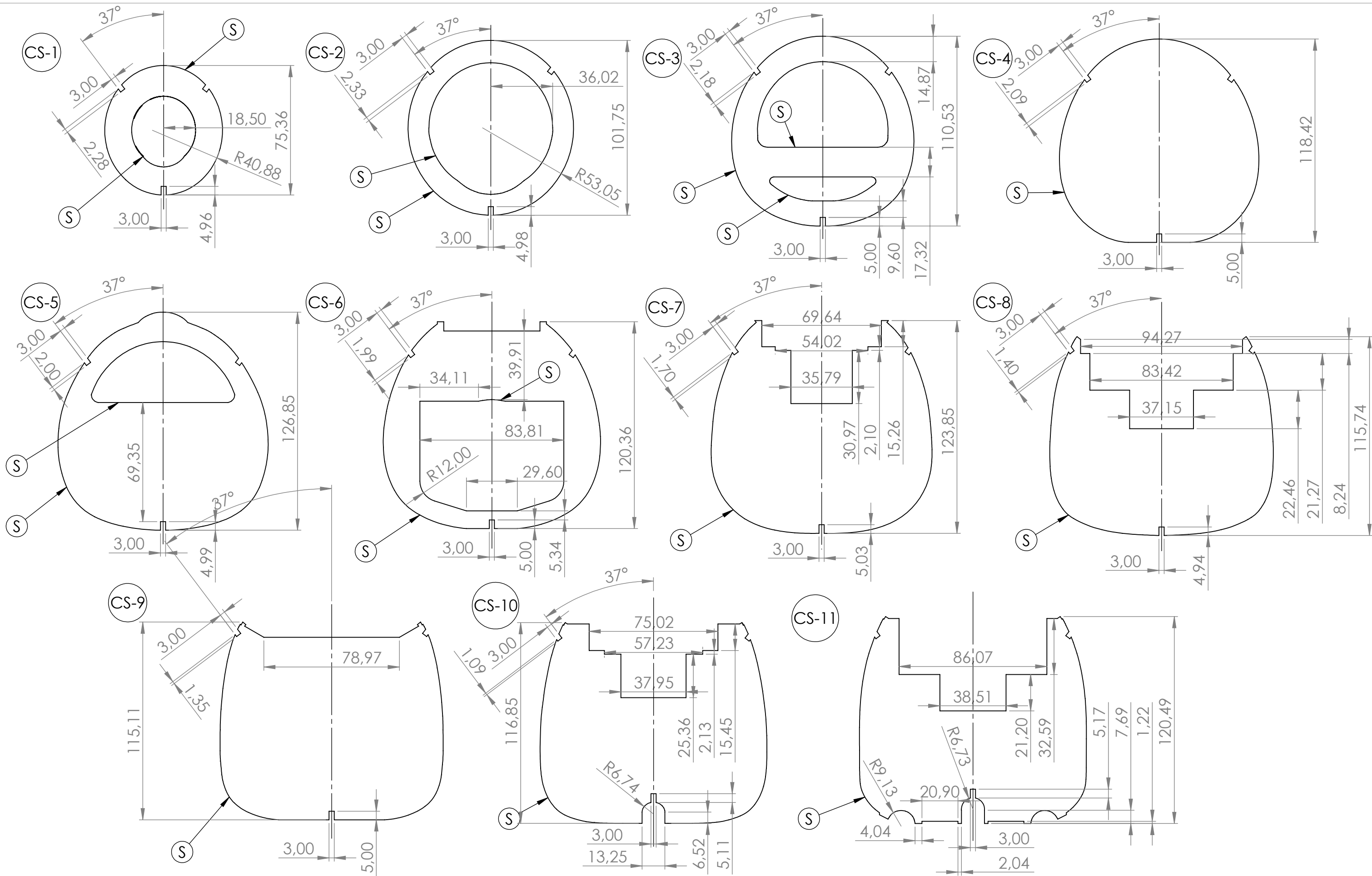
TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:5	Date: 21/06/2023	Size: A3
Sheet: 2	Name: Sabina Campos i Fuentes	UPC - ESEIAAT



TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:6	Date: 21/06/2023	Size: A3
Sheet: 3	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

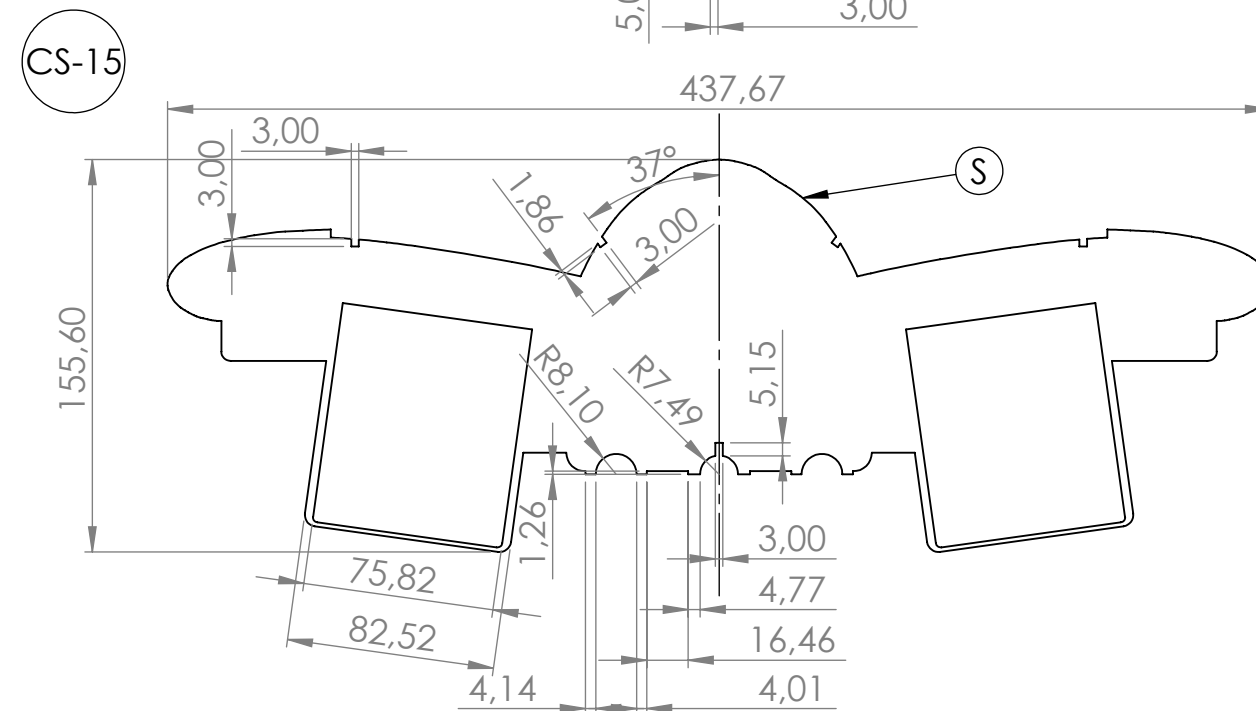
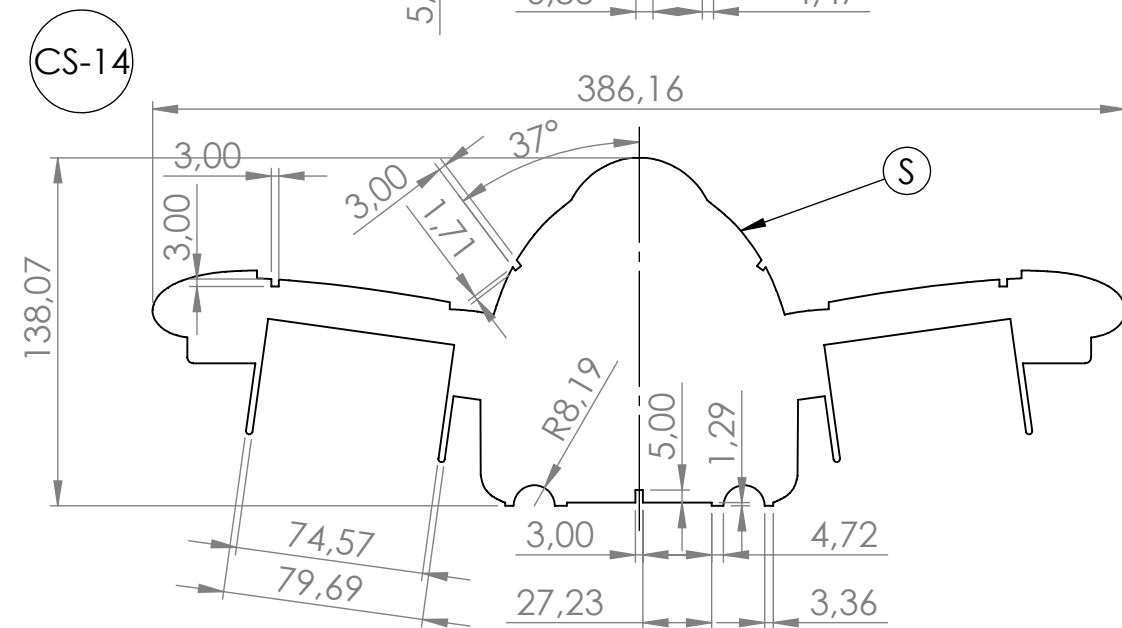
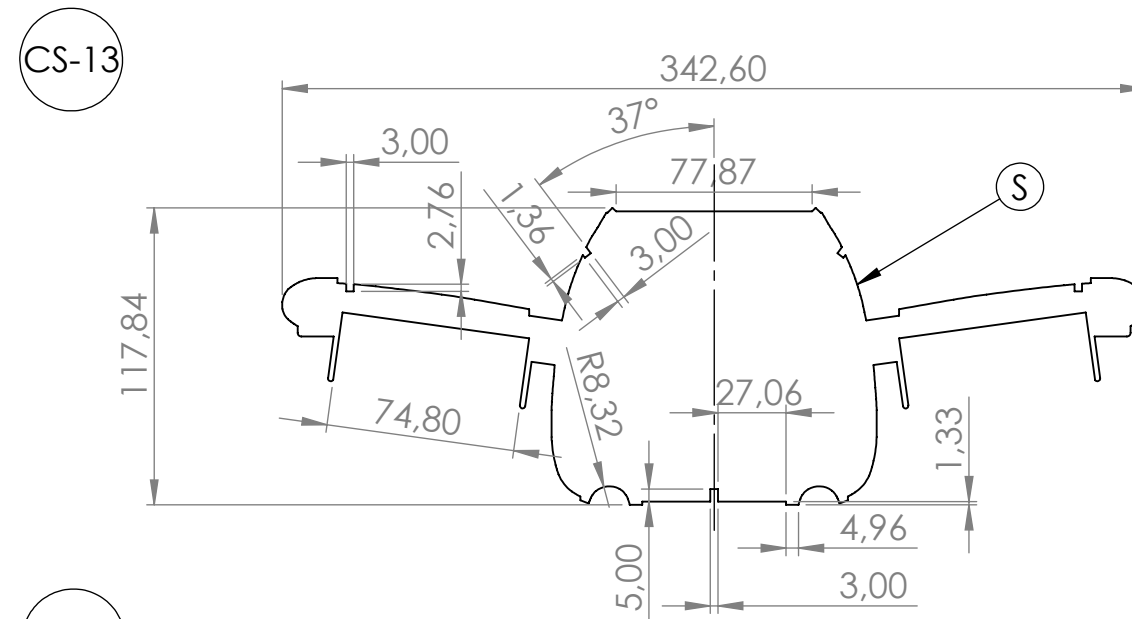
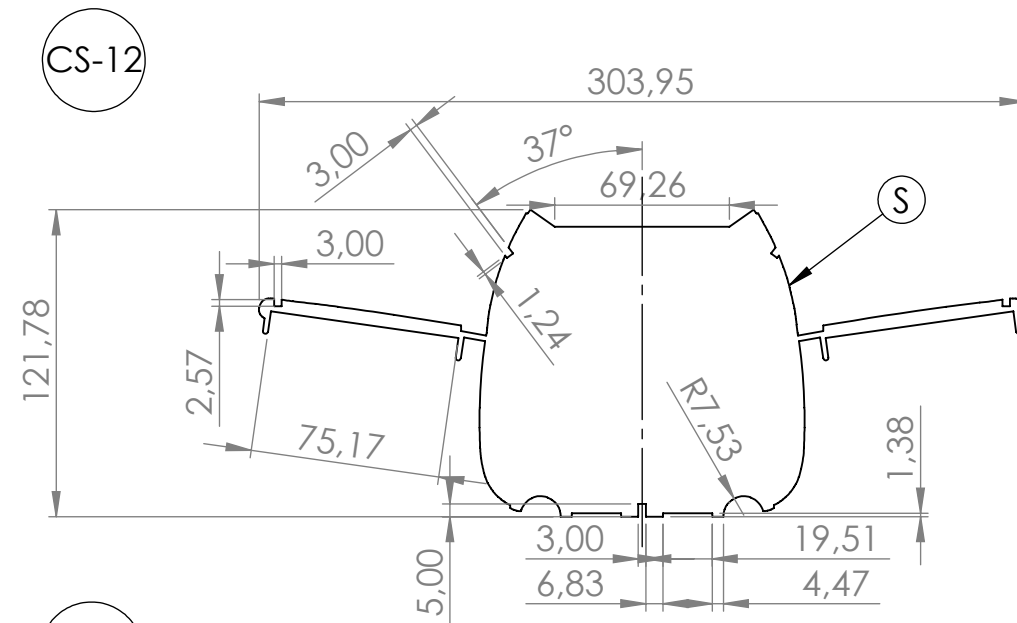


TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:4	Date: 21/06/2023	Size: A3
Sheet: 4	Name: Sabina Campos i Fuentes	UPC - ESEIAAT



All cross-sections corresponding to the fuselage are symmetrical. The axis of symmetry is indicated in the drawing. Only one piece of each will be needed for the scale model. Most of the lines of each piece have been made from splines as indicated in the report, this is indicated in the drawing with a balloon and an S inside. This means that the measurements given do not completely define the part.

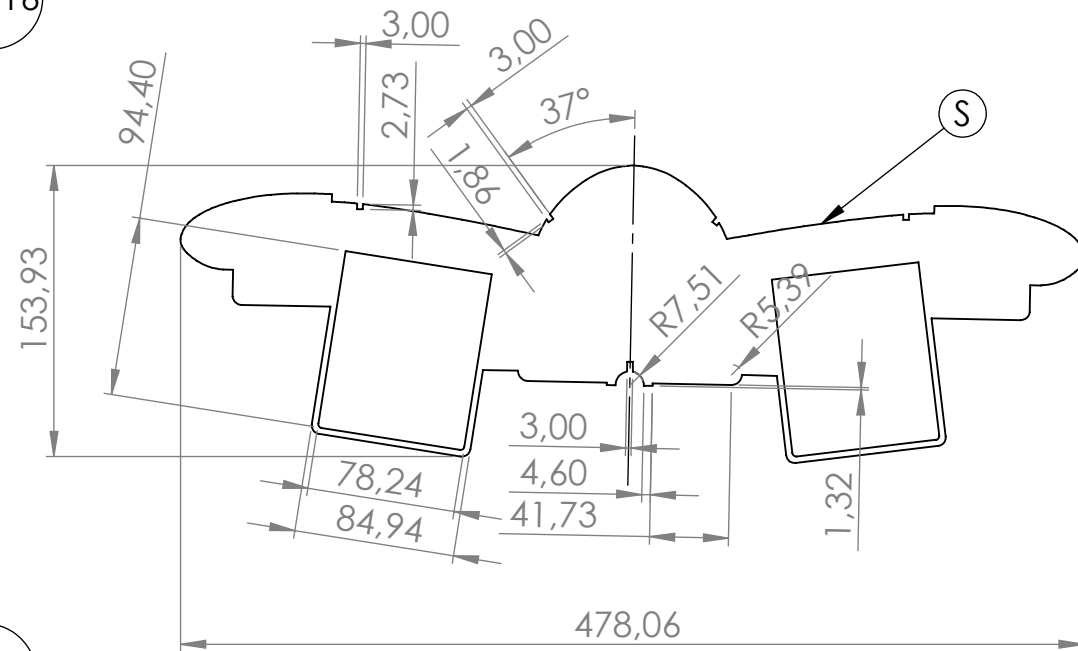
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Scale: 1:2	Date: 21/06/2023	Size: A3
Sheet: 5	Name: Sabina Campos i Fuentes	UPC - ESEIAAT



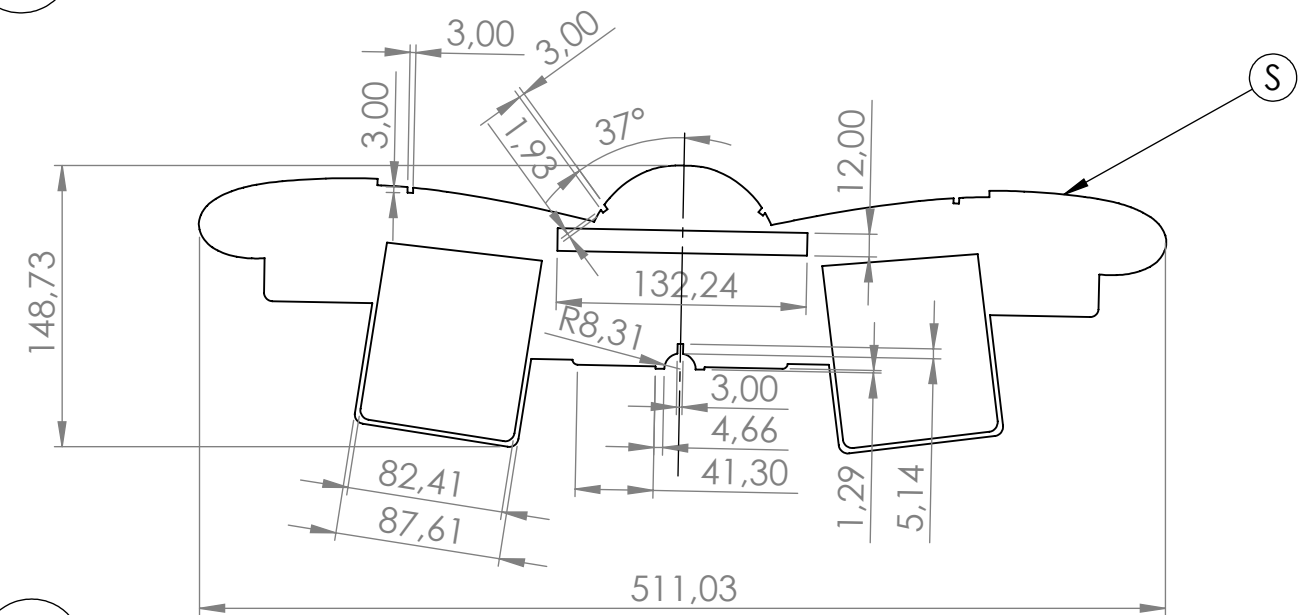
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TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:3	Date: 21/06/2023	Size: A3
Sheet: 6	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

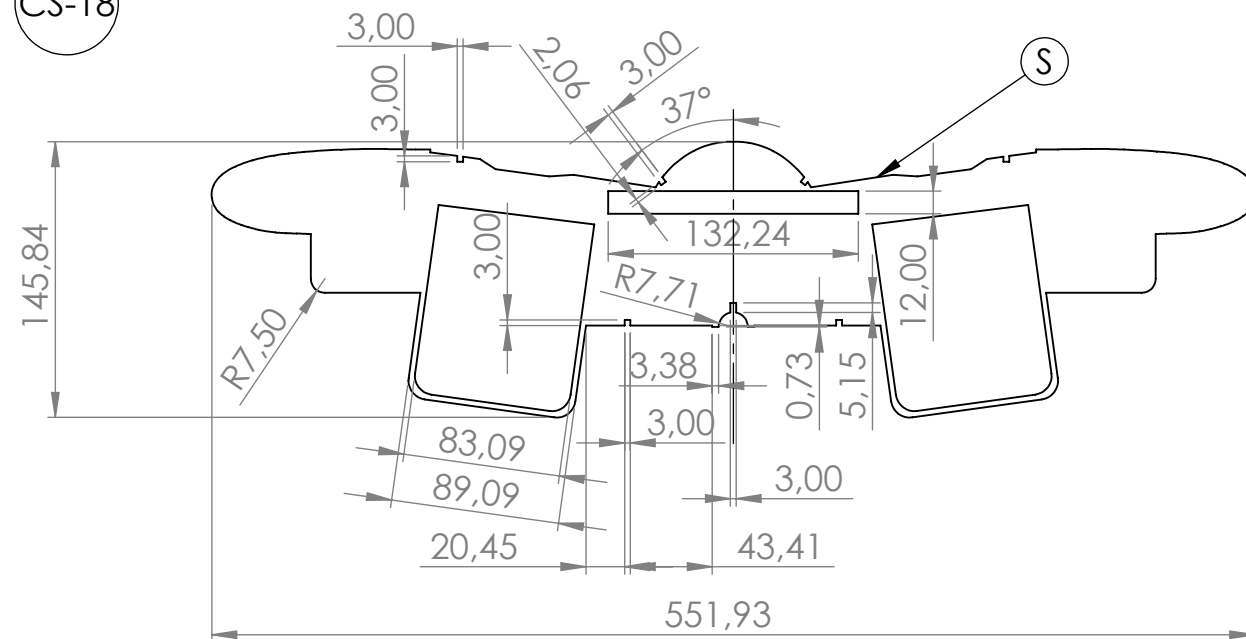
CS-16



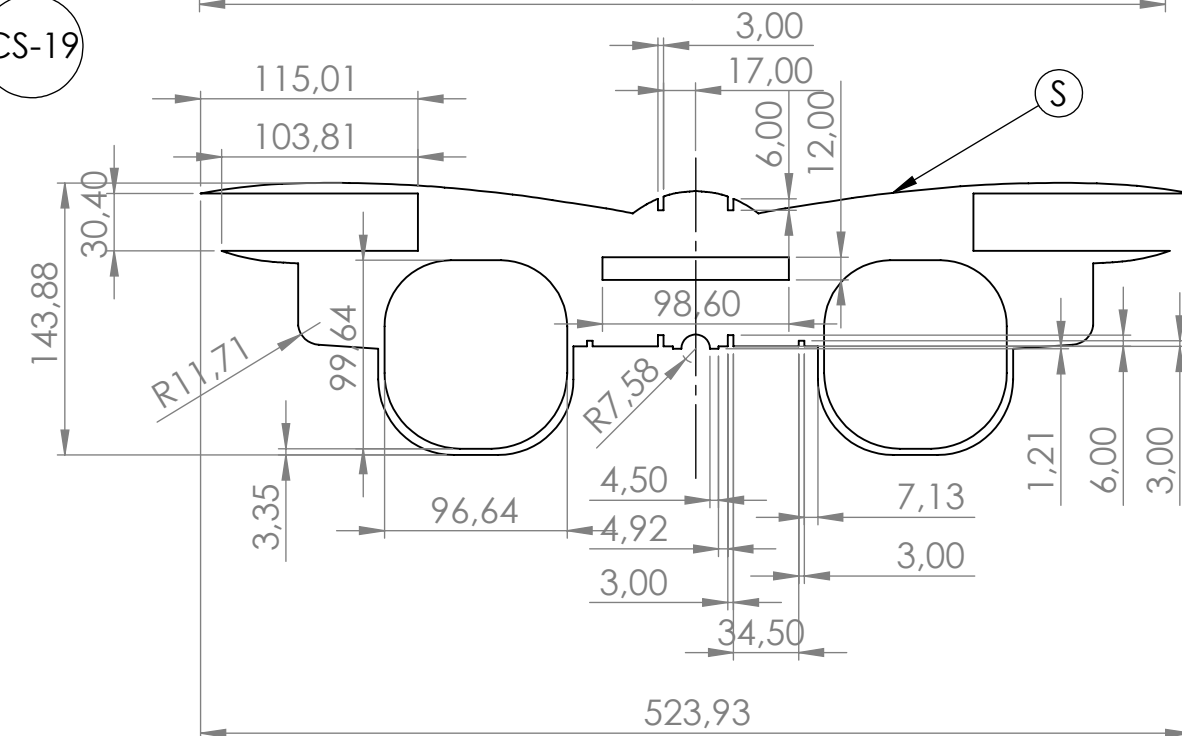
CS-17



CS-18

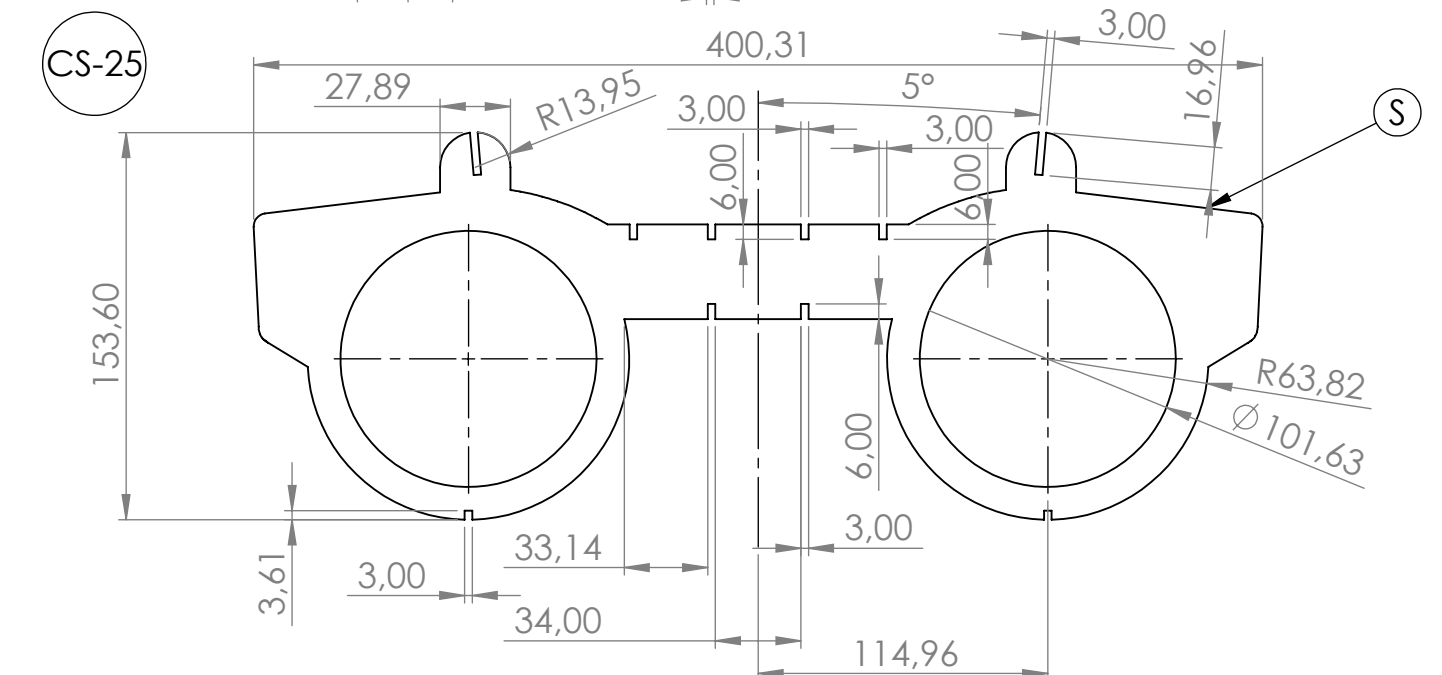
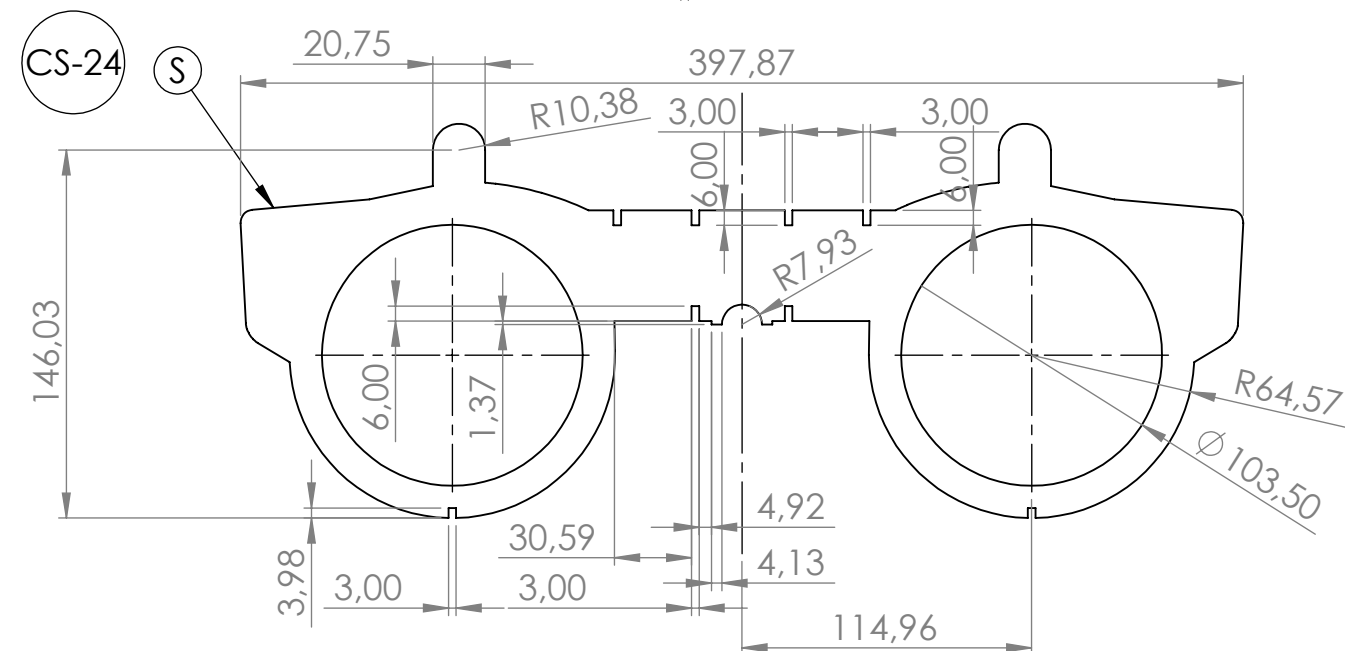
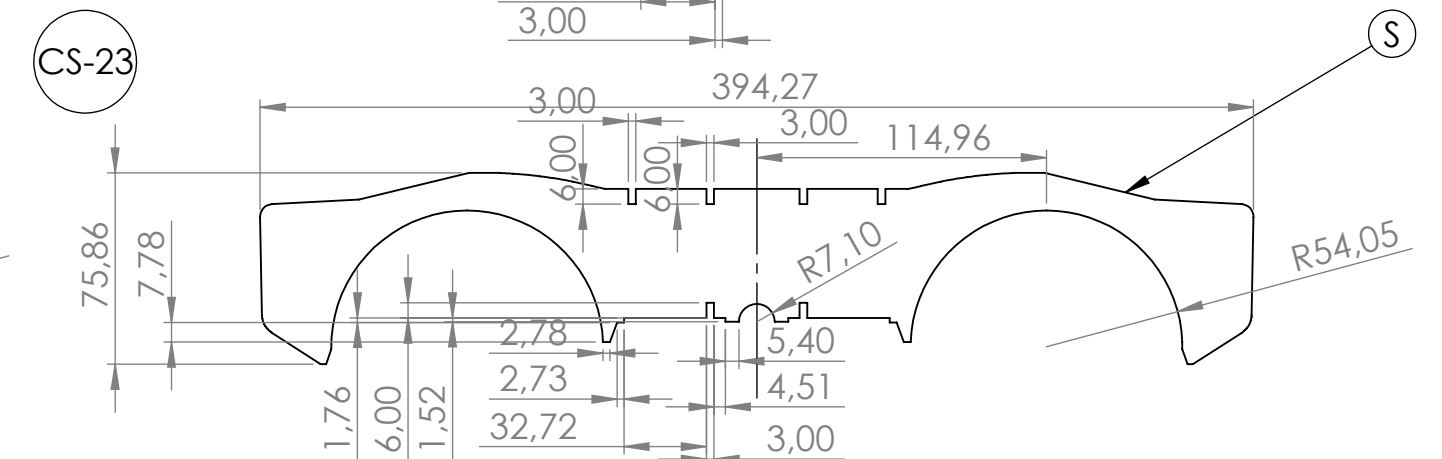
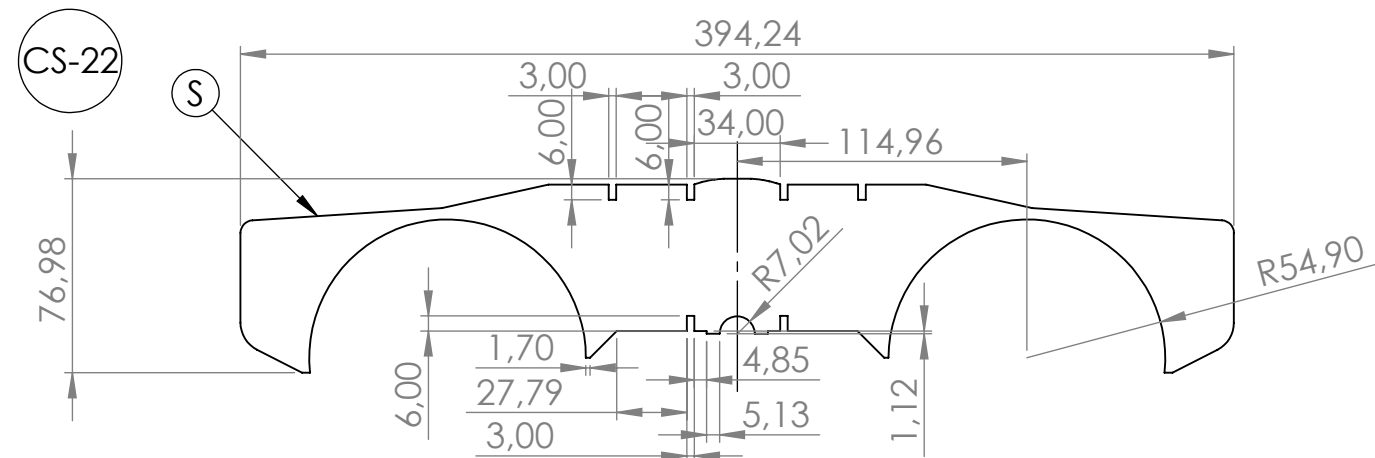
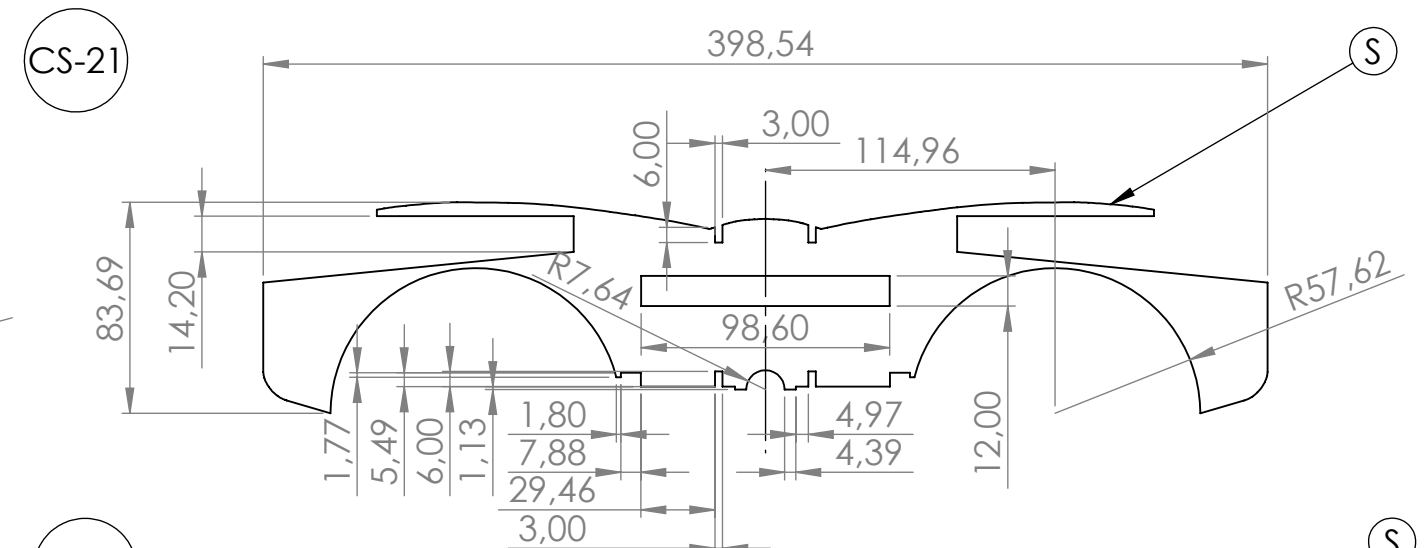
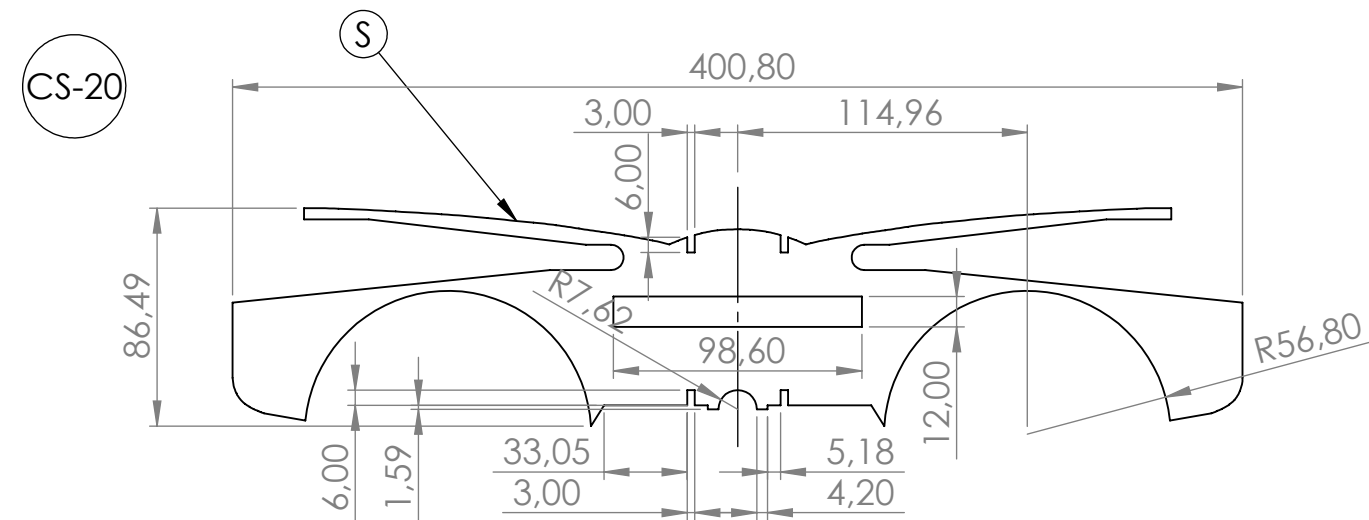


CS-19



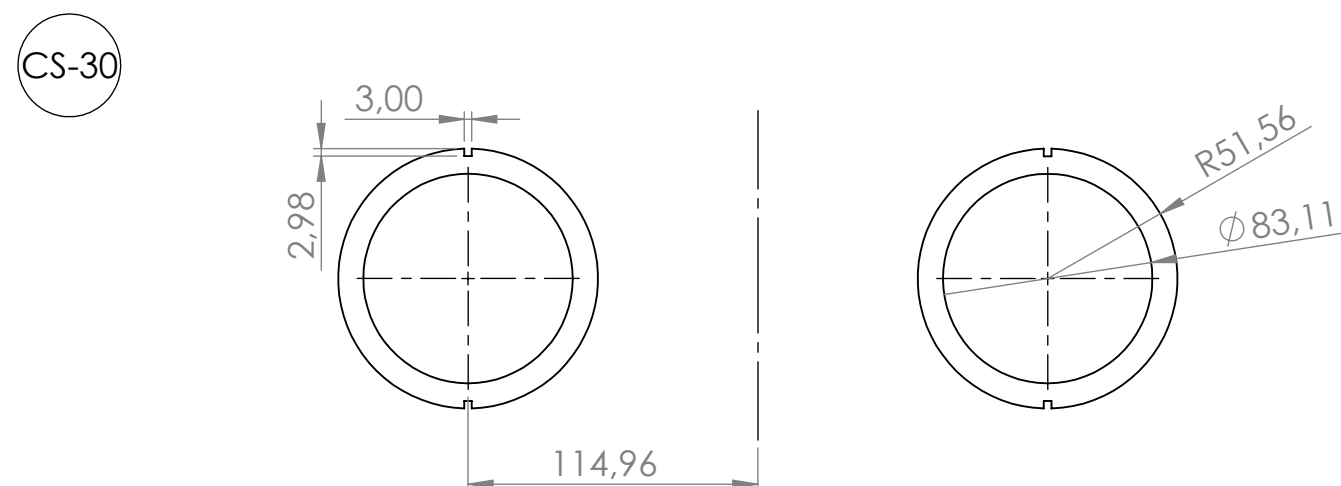
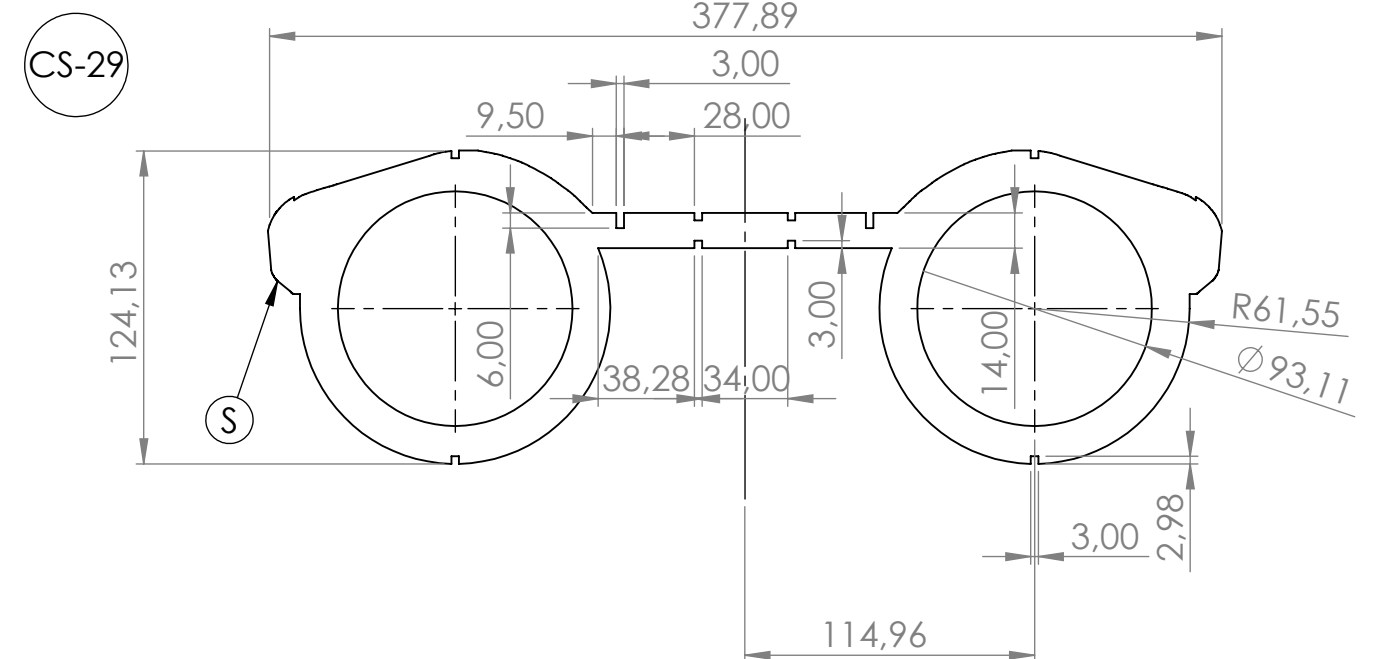
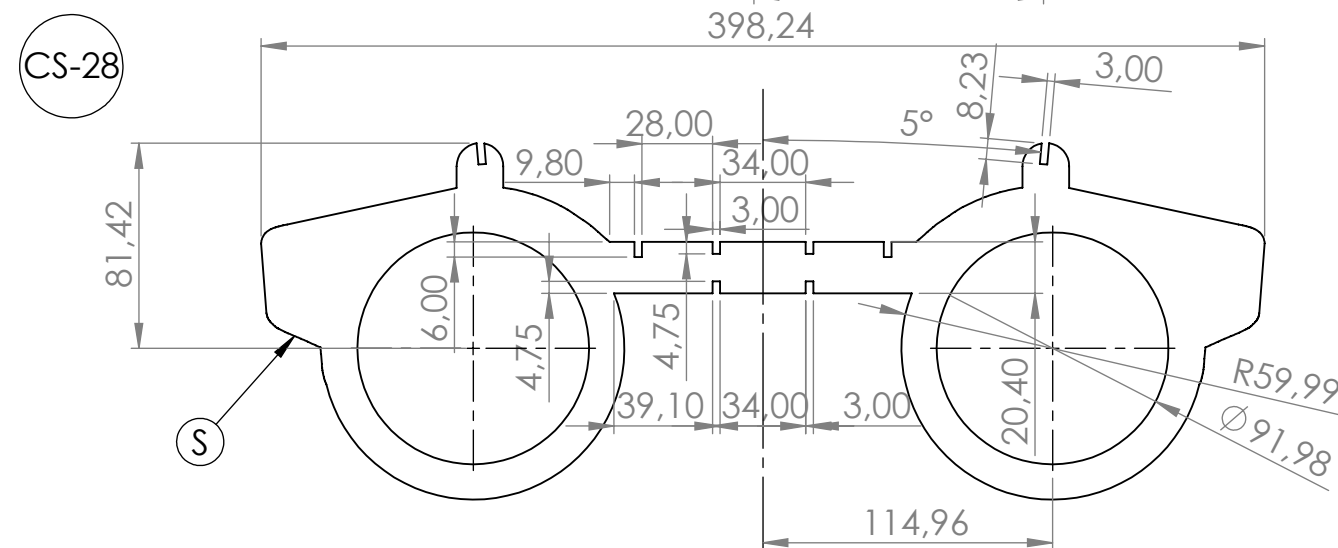
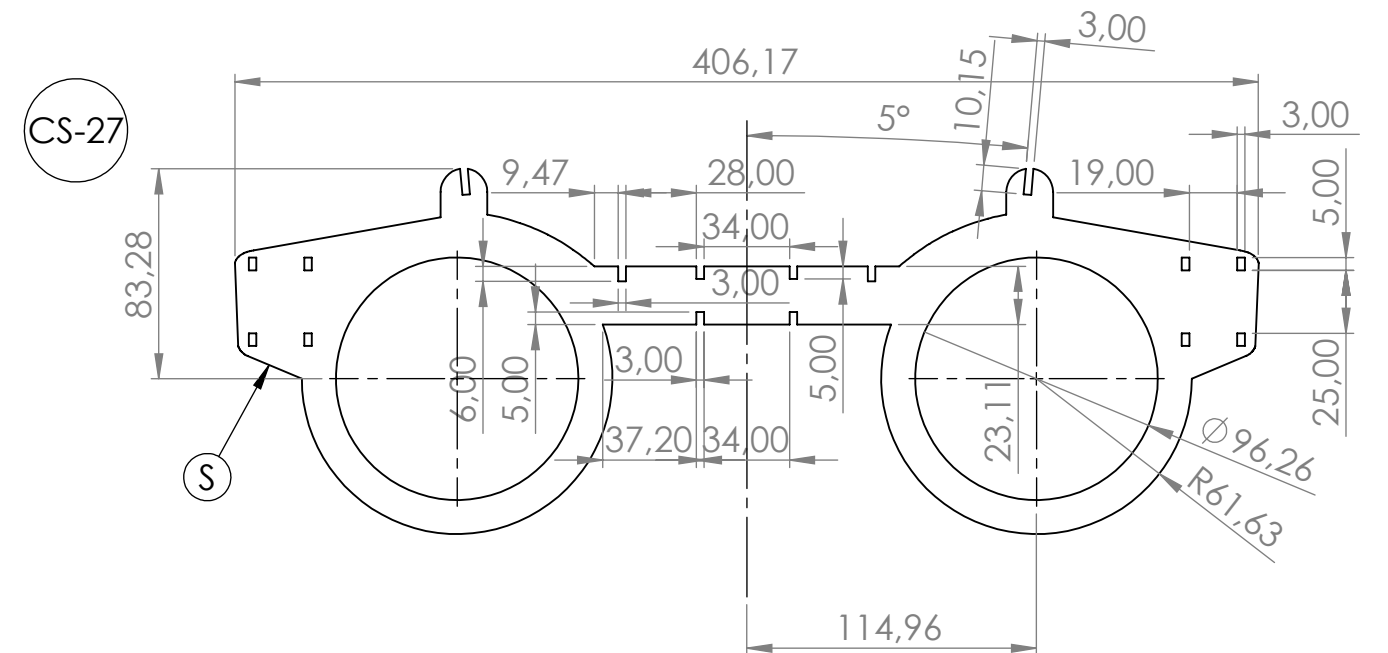
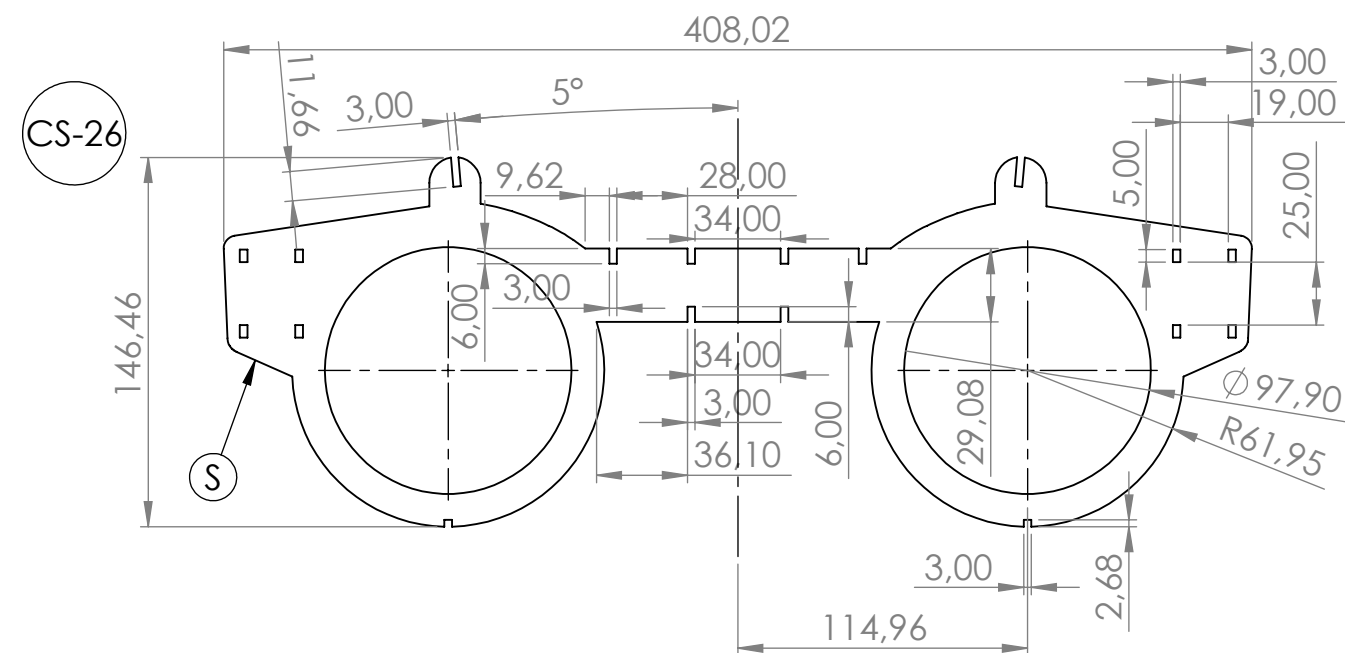
All cross-sections corresponding to the fuselage are symmetrical. The axis of symmetry is indicated in the drawing. Only one piece of each will be needed for the scale model. Most of the lines of each piece have been made from splines as indicated in the report, this is indicated in the drawing with a balloon and an S inside. This means that the measurements given do not completely define the part.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:4	Date: 21/06/2023	Size: A3
Sheet: 7	Name: Sabina Campos i Fuentes	UPC - ESEIAAT



All cross-sections corresponding to the fuselage are symmetrical. The axis of symmetry is indicated in the drawing. Only one piece of each will be needed for the scale model. Most of the lines of each piece have been made from splines as indicated in the report, this is indicated in the drawing with a balloon and an S inside. This means that the measurements given do not completely define the part.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:3	Date: 21/06/2023	Size: A3
Sheet: 8	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

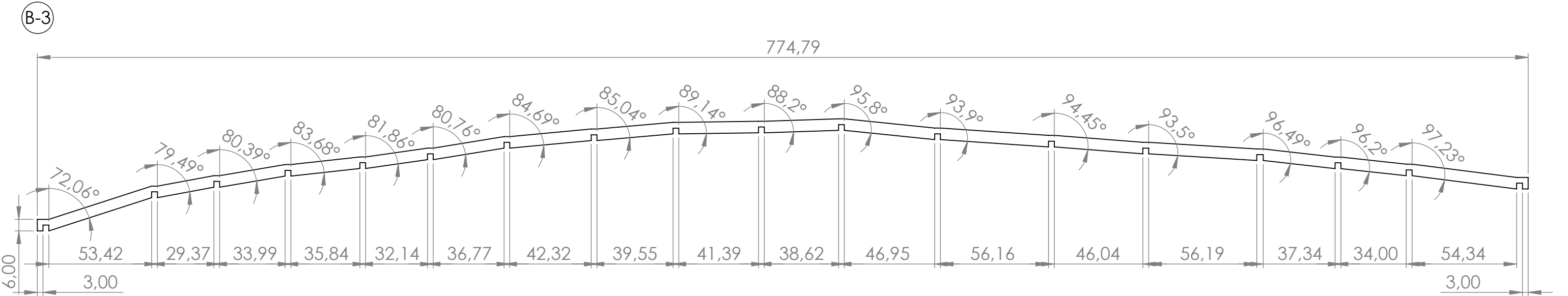
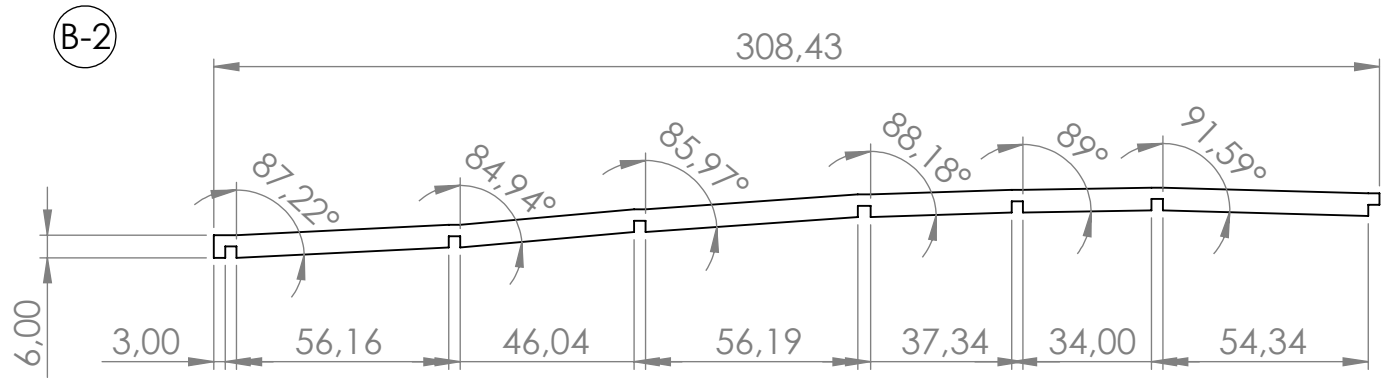
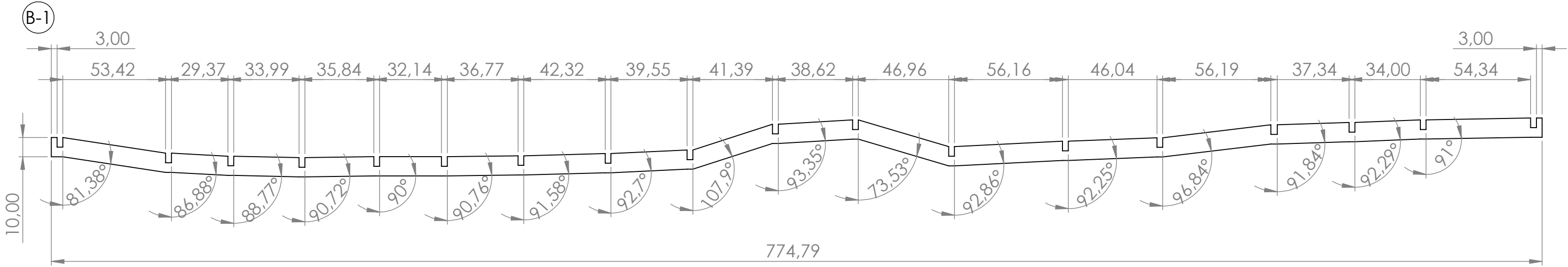


All cross-sections corresponding to the fuselage are symmetrical. The axis of symmetry is indicated in the drawing. Only one piece of each will be needed for the scale model. Most of the lines of each piece have been made from splines as indicated in the report, this is indicated in the drawing with a balloon and an S inside. This means that the measurements given do not completely define the part.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:3	Date: 21/06/2023	Size: A3
Sheet: 9	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

All the notches of the cross members are 3mm x 3mm. In order not to saturate the drawings, these measures have not been added.

- Beam 1, one needed for the scale model.
- Beam 2, two needed for the scale model.
- Beam 3, two needed for the scale model.

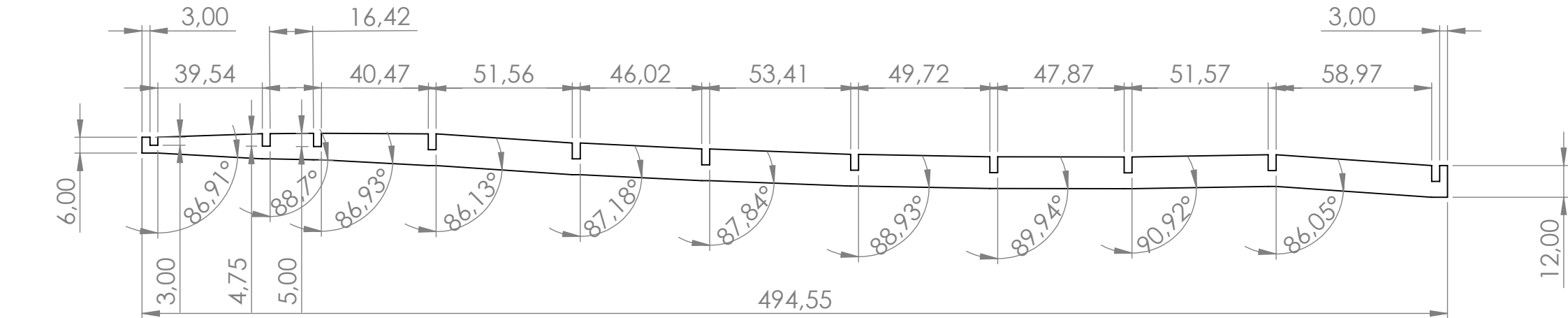


TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:2	Date: 21/06/2023	Size: A3
Sheet: 10	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

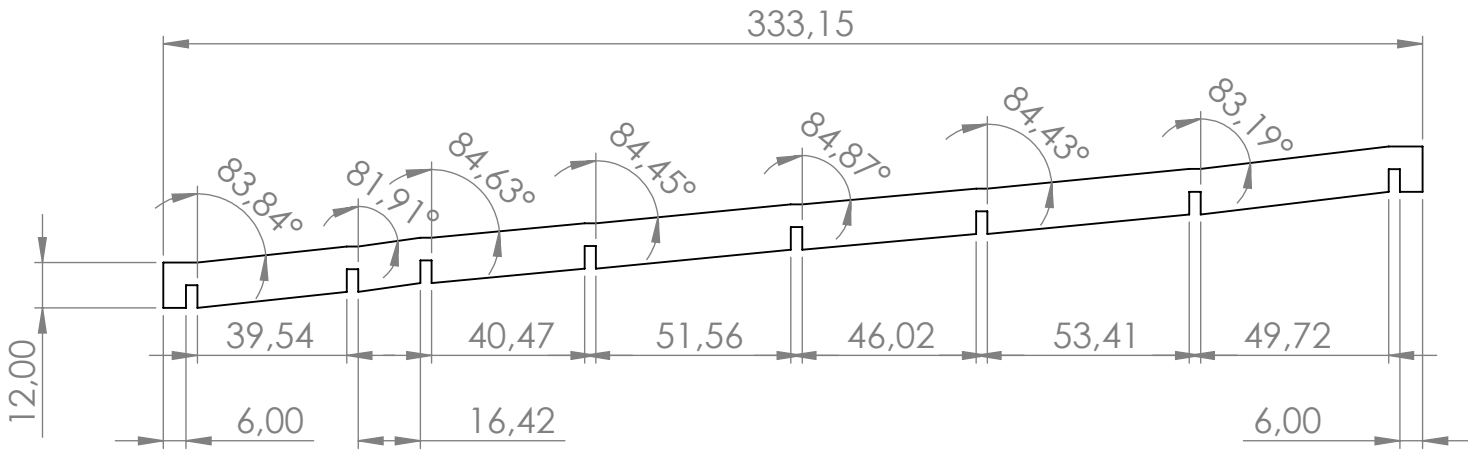
All the notches in crossbeams B-4 are 3mm x 6mm, except for the first three. All the notches in crossbeams B-5 are 3mm x 6mm.
All the notches in crossbeams B-6 and B-7 are 3mm x 3mm. In order not to saturate the drawings, these measures have not been added.

- Beam 4, four needed for the scale model.
- Beam 5, two needed for the scale model.
- Beam 6, two needed for the scale model.
- Beam 7, four needed for the scale model.

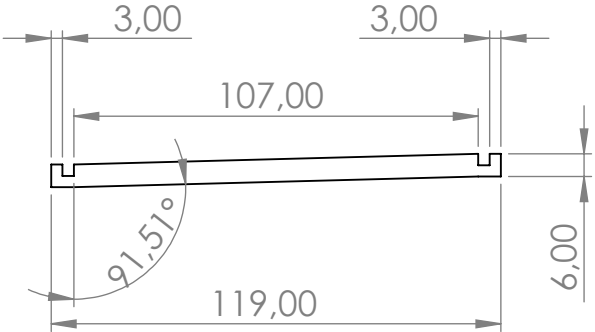
B-4



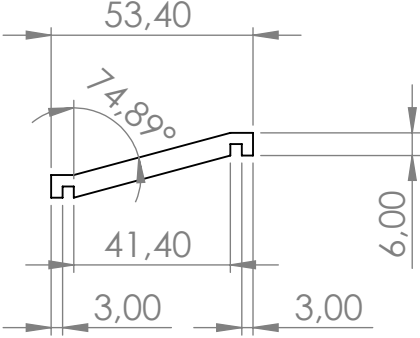
B-5



B-6

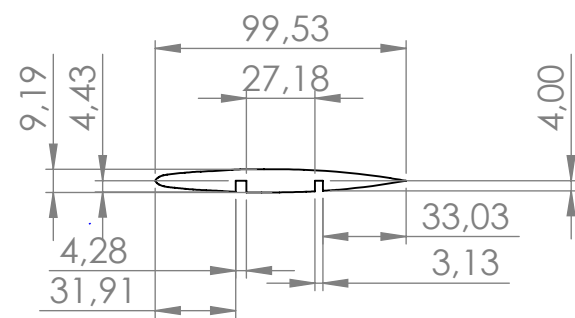


B-7

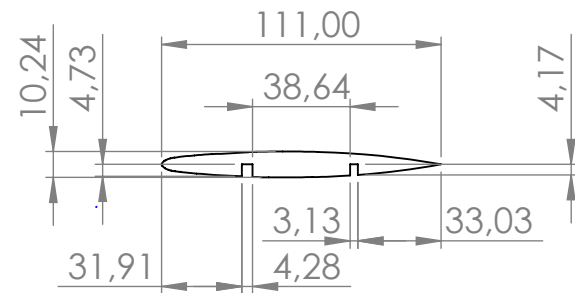


TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:2	Date: 21/06/2023	Size: A3
Sheet: 11	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

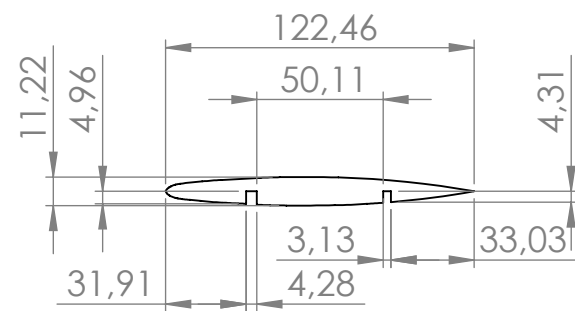
R-1



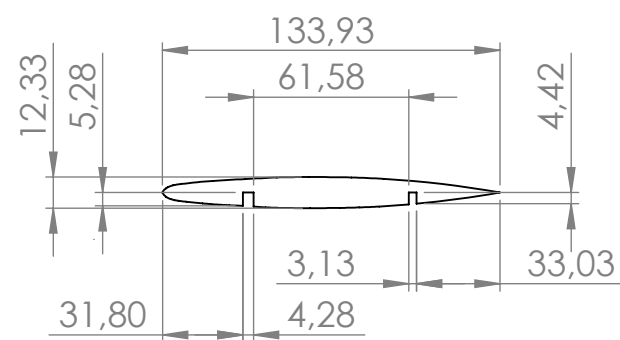
R-2



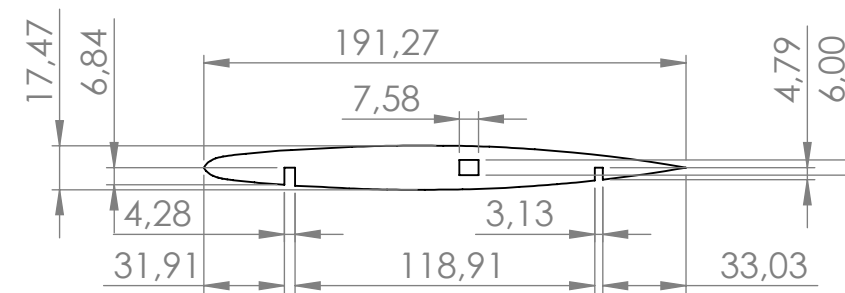
R-3



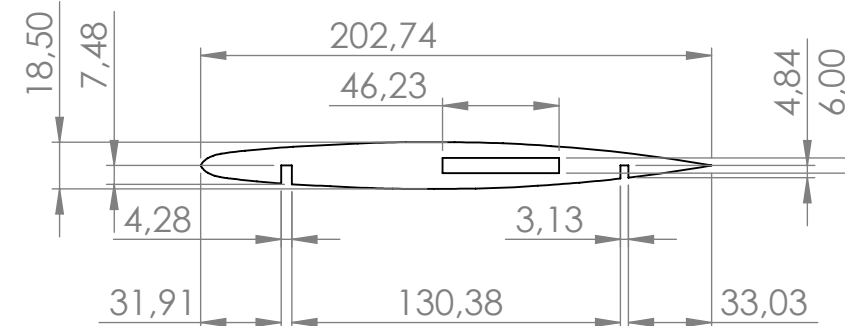
R-4



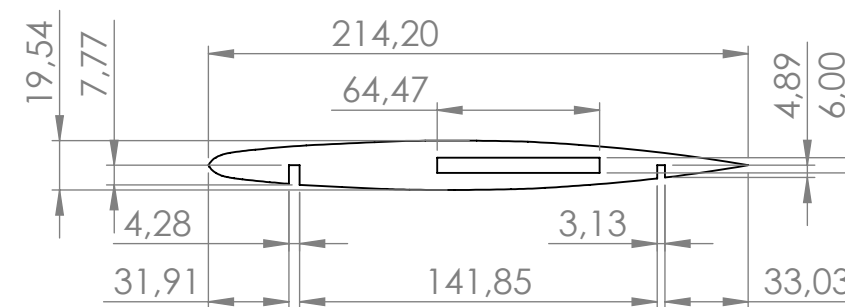
R-5



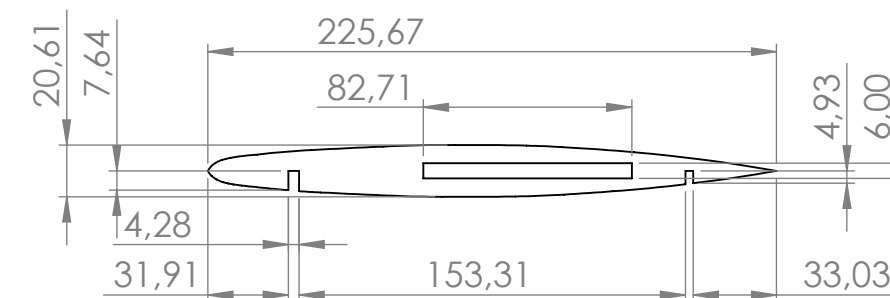
R-6



R-7

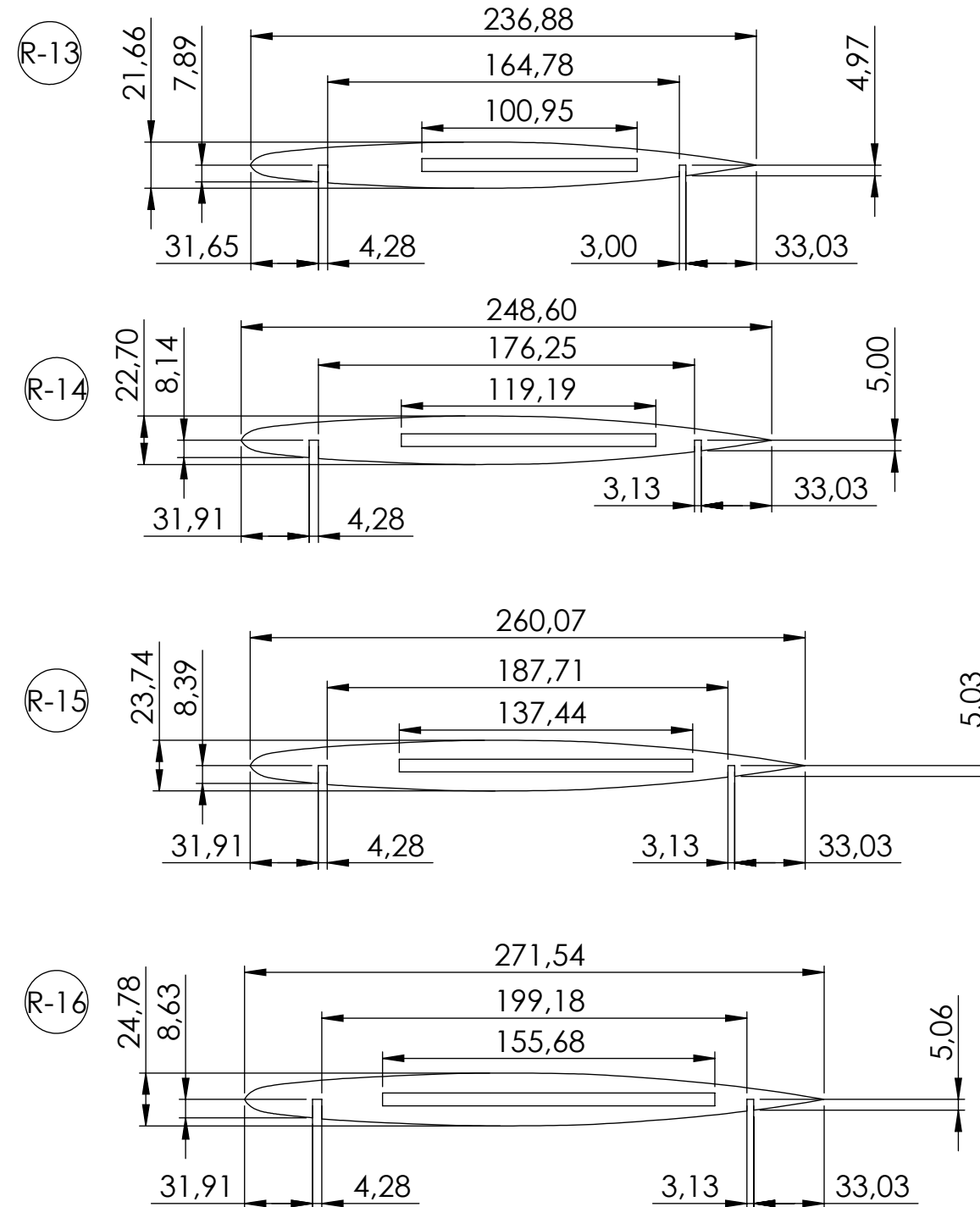
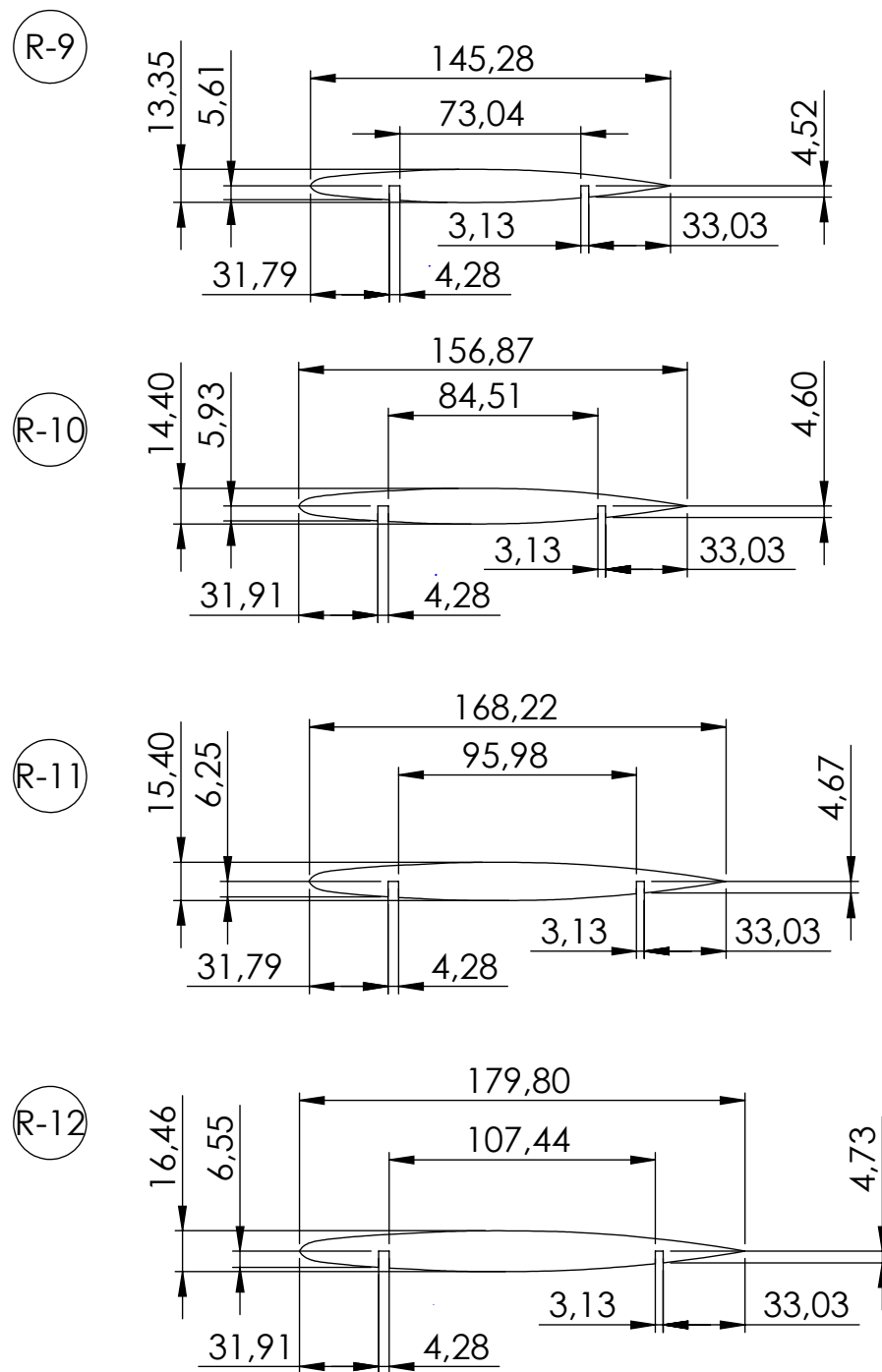


R-8



The wing has ribs with a NACA airfoil 64A209.65 at the root and 64A208.91 at the tip.

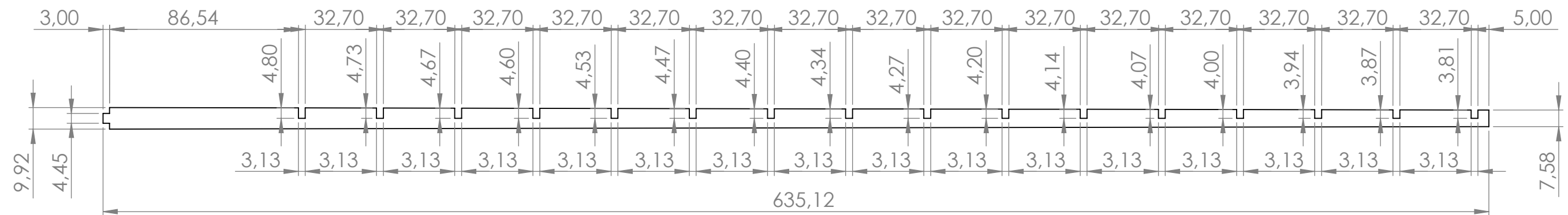
TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:3	Date: 21/06/2023	Size: A3
Sheet: 12	Name: Sabina Campos i Fuentes	UPC - ESEIAAT



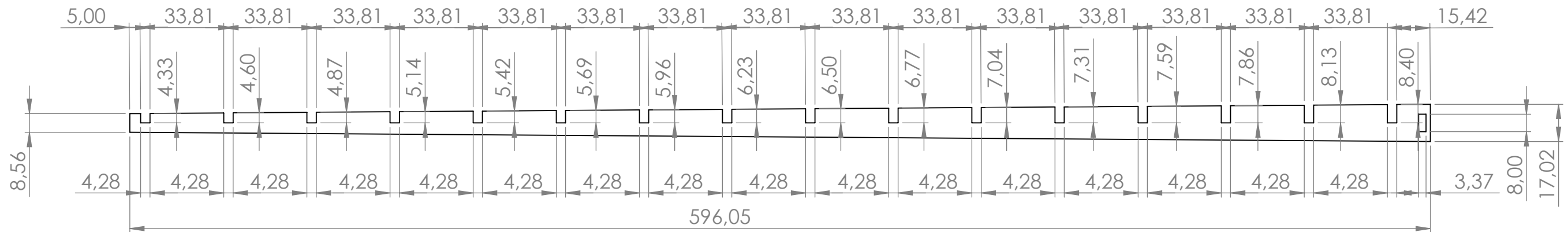
The wing has ribs with a NACA airfoil 64A209.65 at the root and 64A208.91 at the tip.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:3	Date: 21/06/2023	Size: A3
Sheet: 13	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

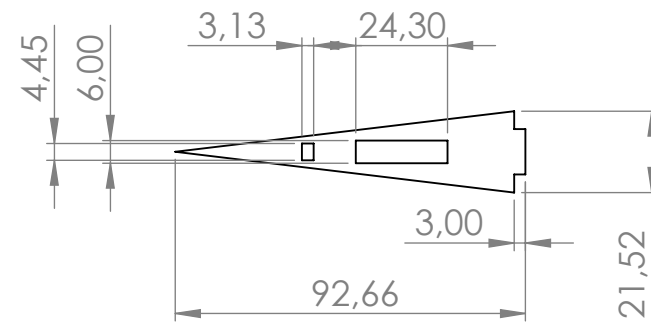
WB-1



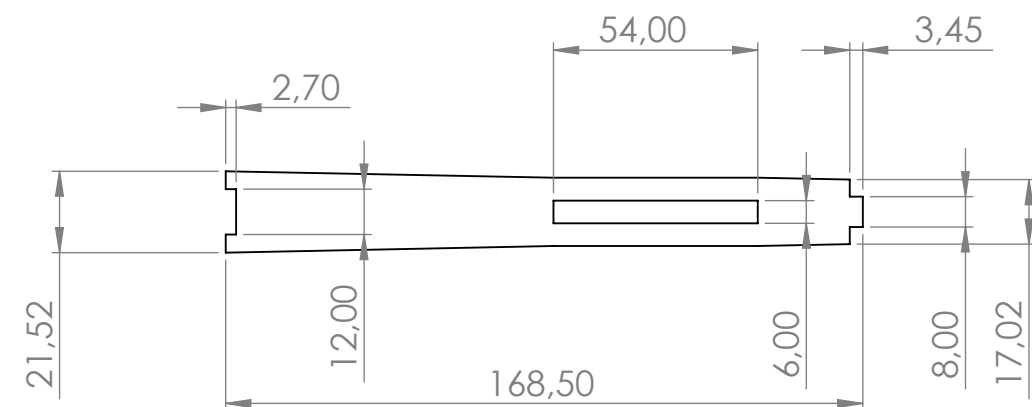
WB-2



WC-1



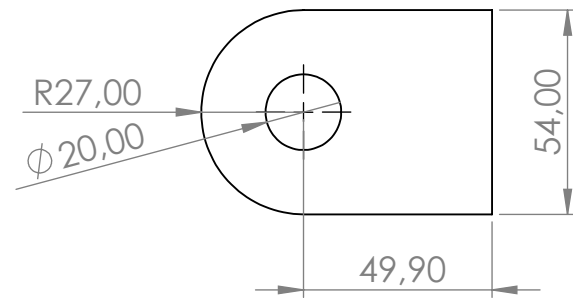
WC-2



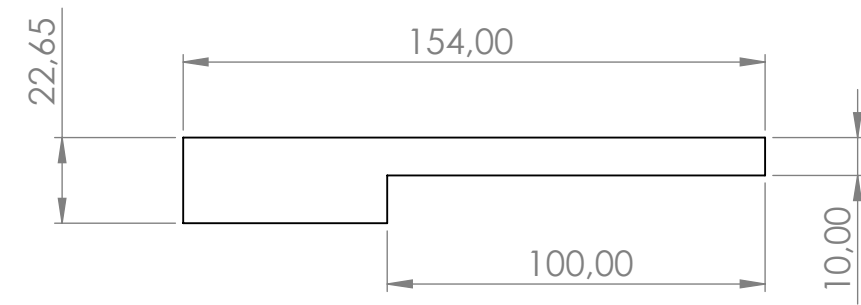
On this page we can see the wing beams and their connectors. These parts are completely defined.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:2	Date: 21/06/2023	Size: A3
Sheet: 14	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

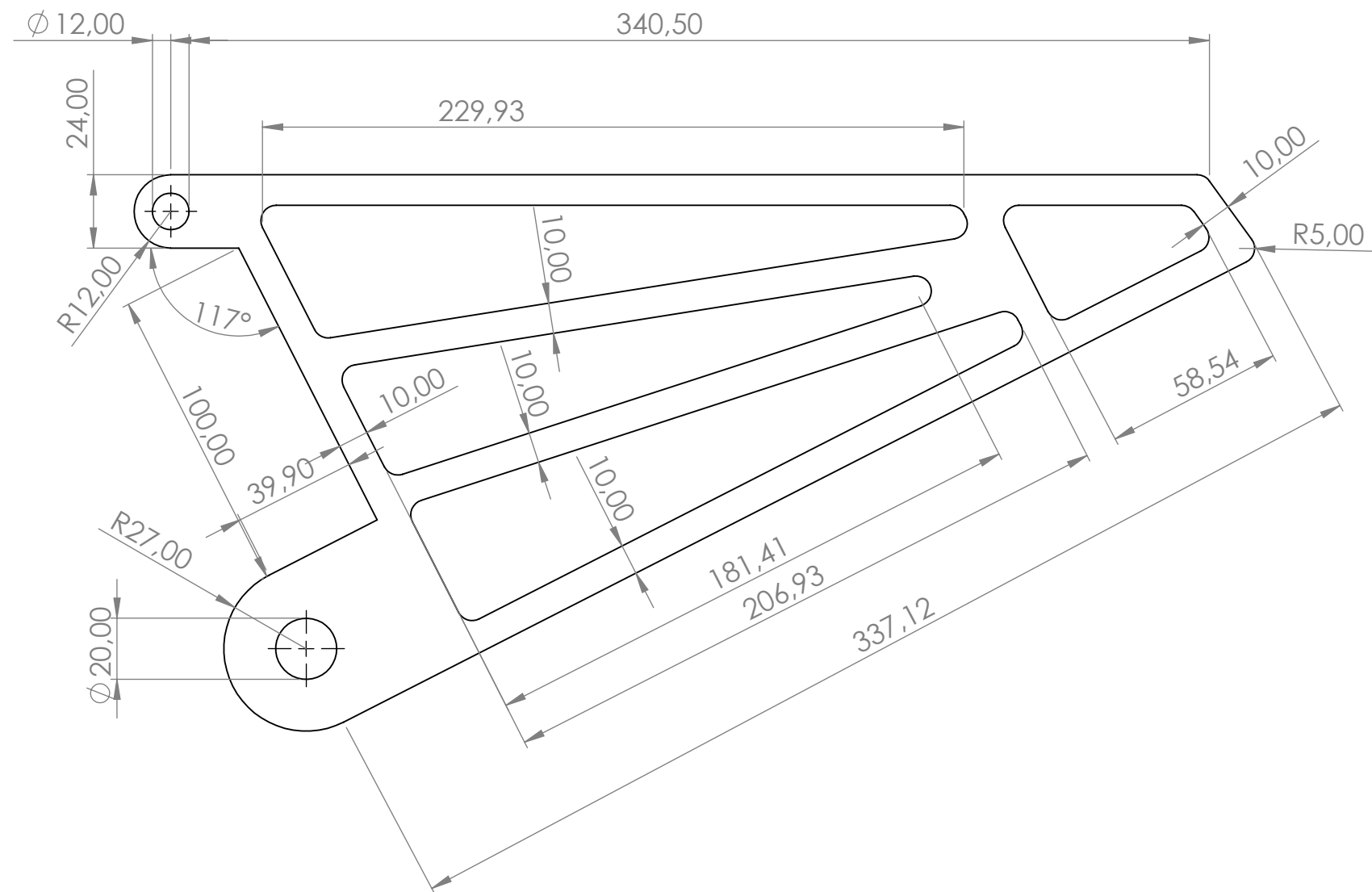
WS-1



WS-2



WS-3

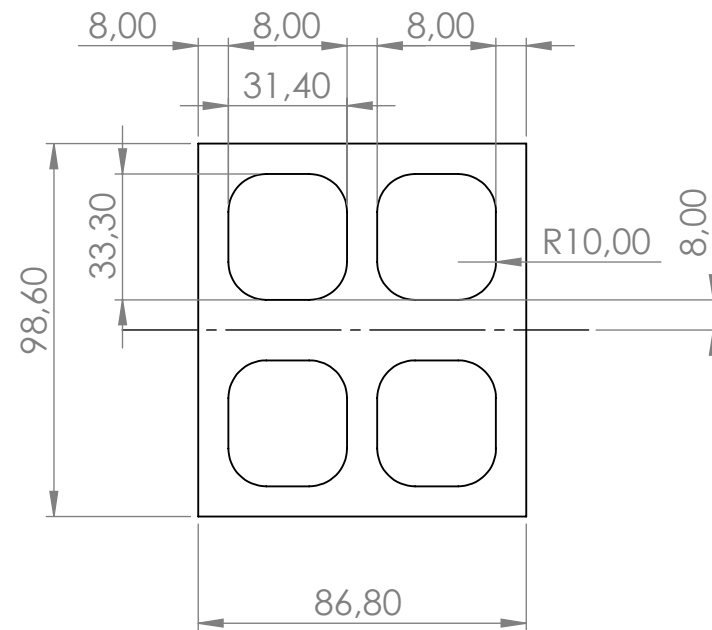


All roundings of the parts are R5, they have not been added to the plan in order not to clutter it.

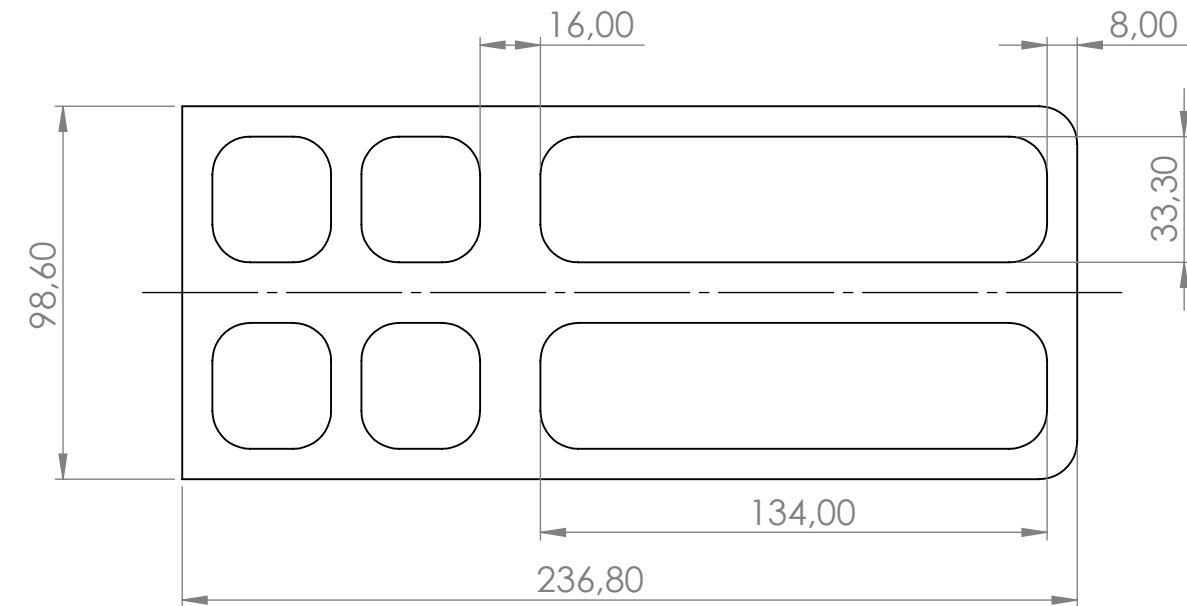
- WS-1, eight needed for the scale model.
- WS-2, eight needed for the scale model.
- WS-3, four needed for the scale model.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:2	Date: 21/06/2023	Size: A3
Sheet: 15	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

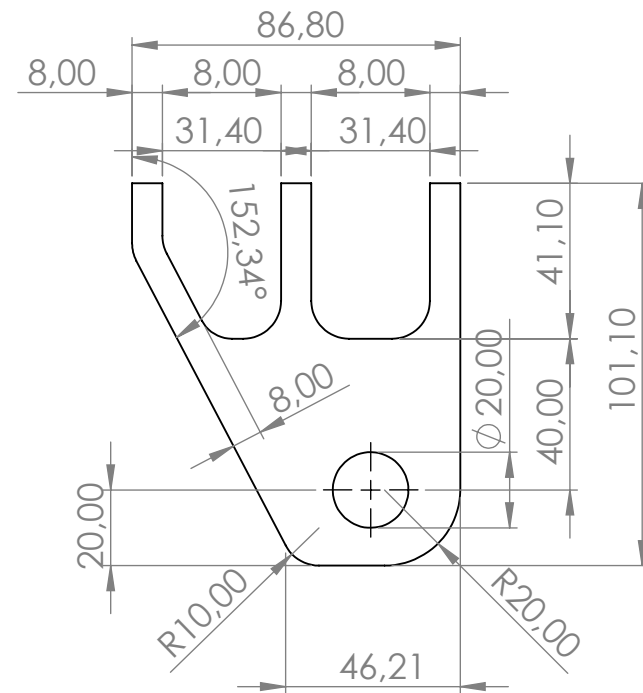
PM-1



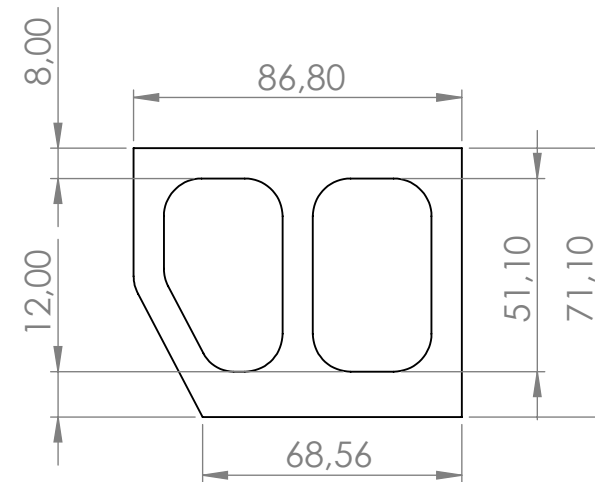
PM-2



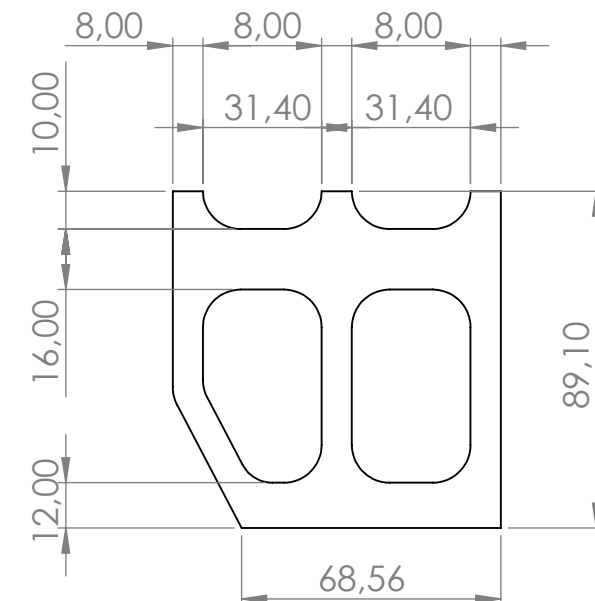
PM-3



PM-4



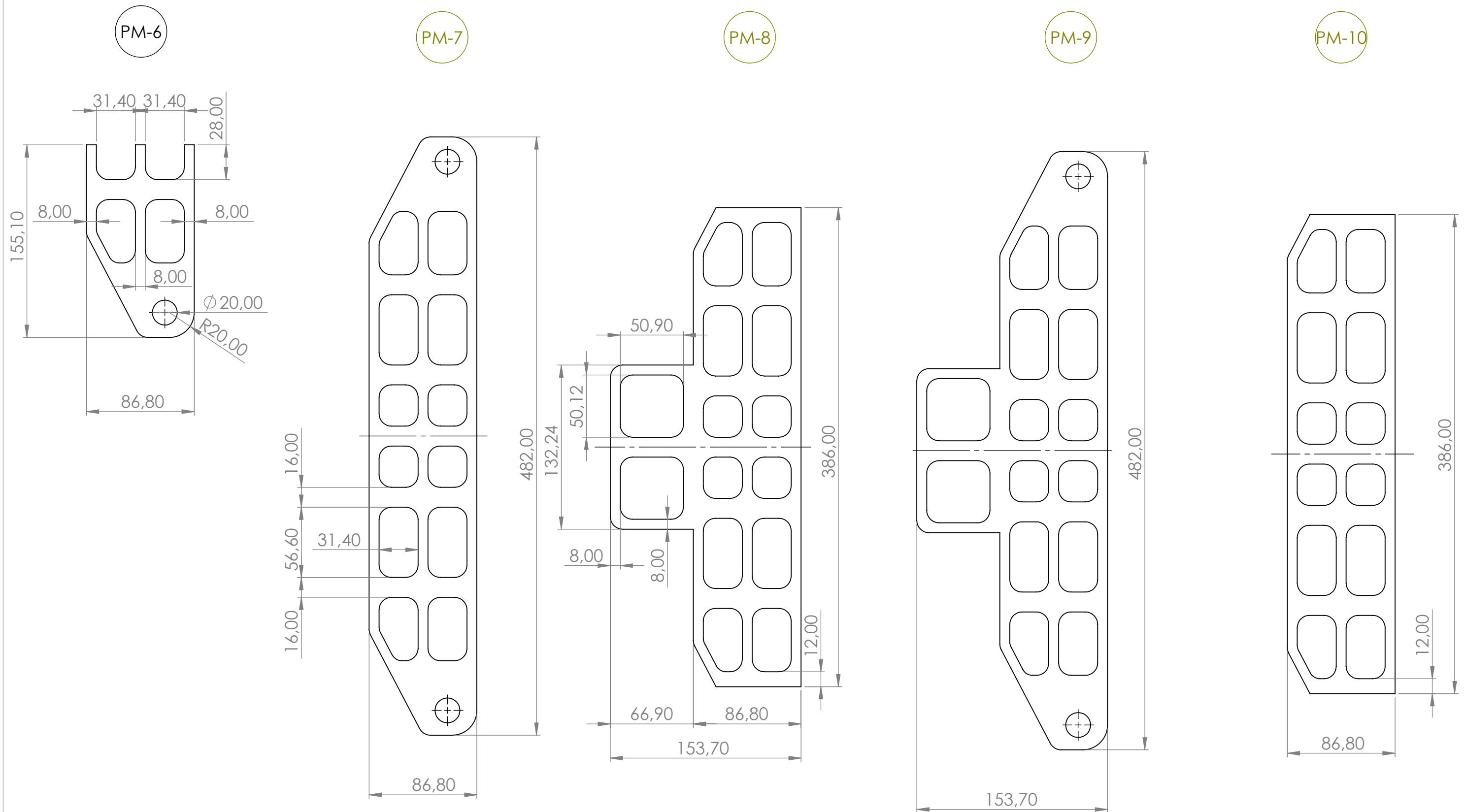
PM-5



All roundings of the parts are R10, they have not been added to the drawings in order not to clutter it. As these parts are mounted on top of each other, they share most of the same dimensions. For this reason, if some of these measures have already been indicated in another piece, they have not been added in the others.

- PM-1, two needed for the scale model.
- PM-2, four needed for the scale model.
- PM-3, two needed for the scale model.
- PM-4, two needed for the scale model.
- PM-5, two needed for the scale model.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:2	Date: 21/06/2023	Size: A3
Sheet: 16	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

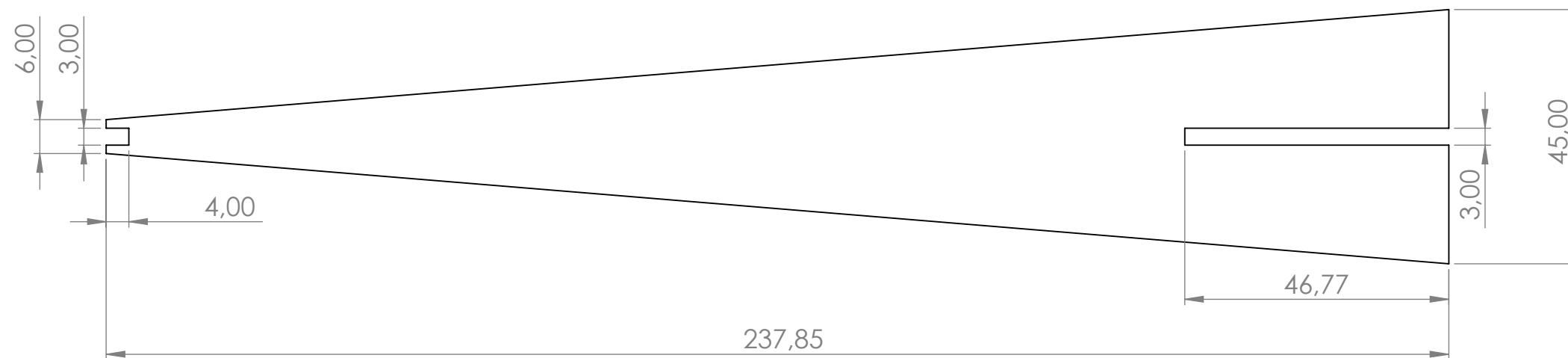


All roundings of the parts are R10, they have not been added to the drawings in order not to clutter it. As these parts are mounted on top of each other, they share most of the same dimensions. For this reason, if some of these measures have already been indicated in another piece, they have not been added in the others.

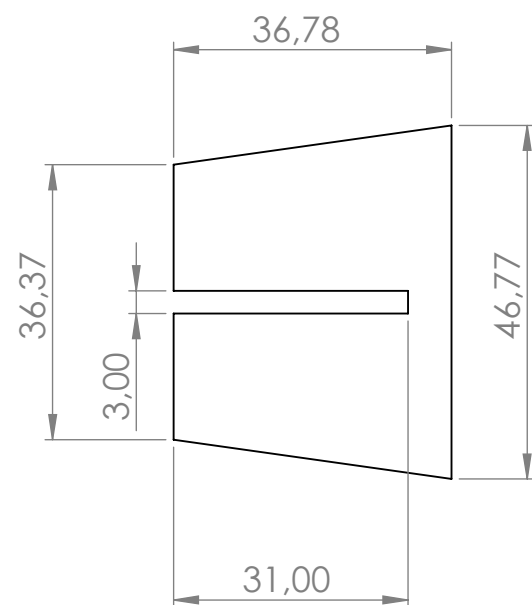
- PM-6, two needed for the scale model.
- PM-7, two needed for the scale model.
- PM-8, two needed for the scale model.
- PM-9, two needed for the scale model.
- PM-10, two needed for the scale model.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:3	Date: 21/06/2023	Size: A3
Sheet: 17	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

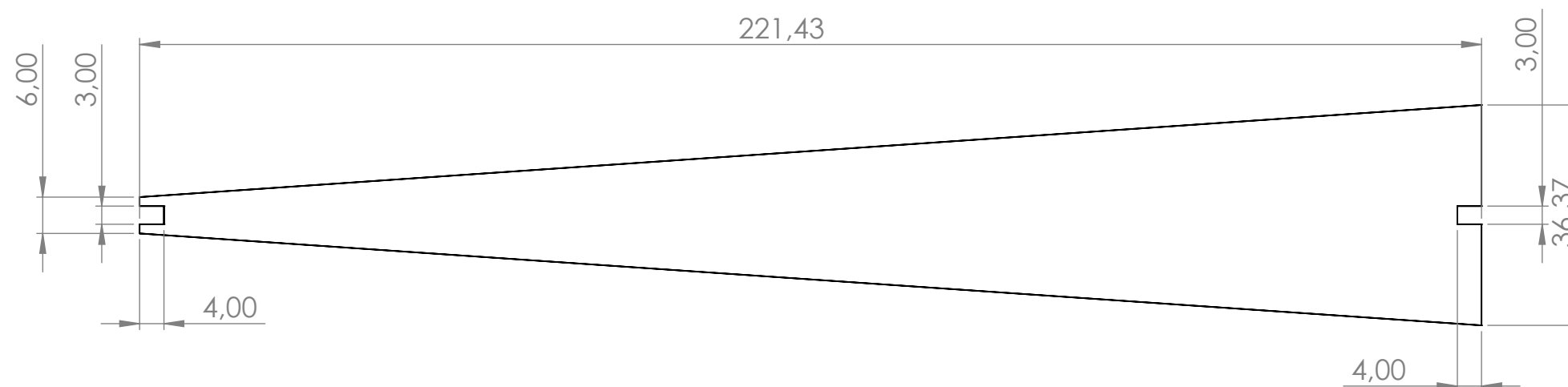
VS-1



VS-2



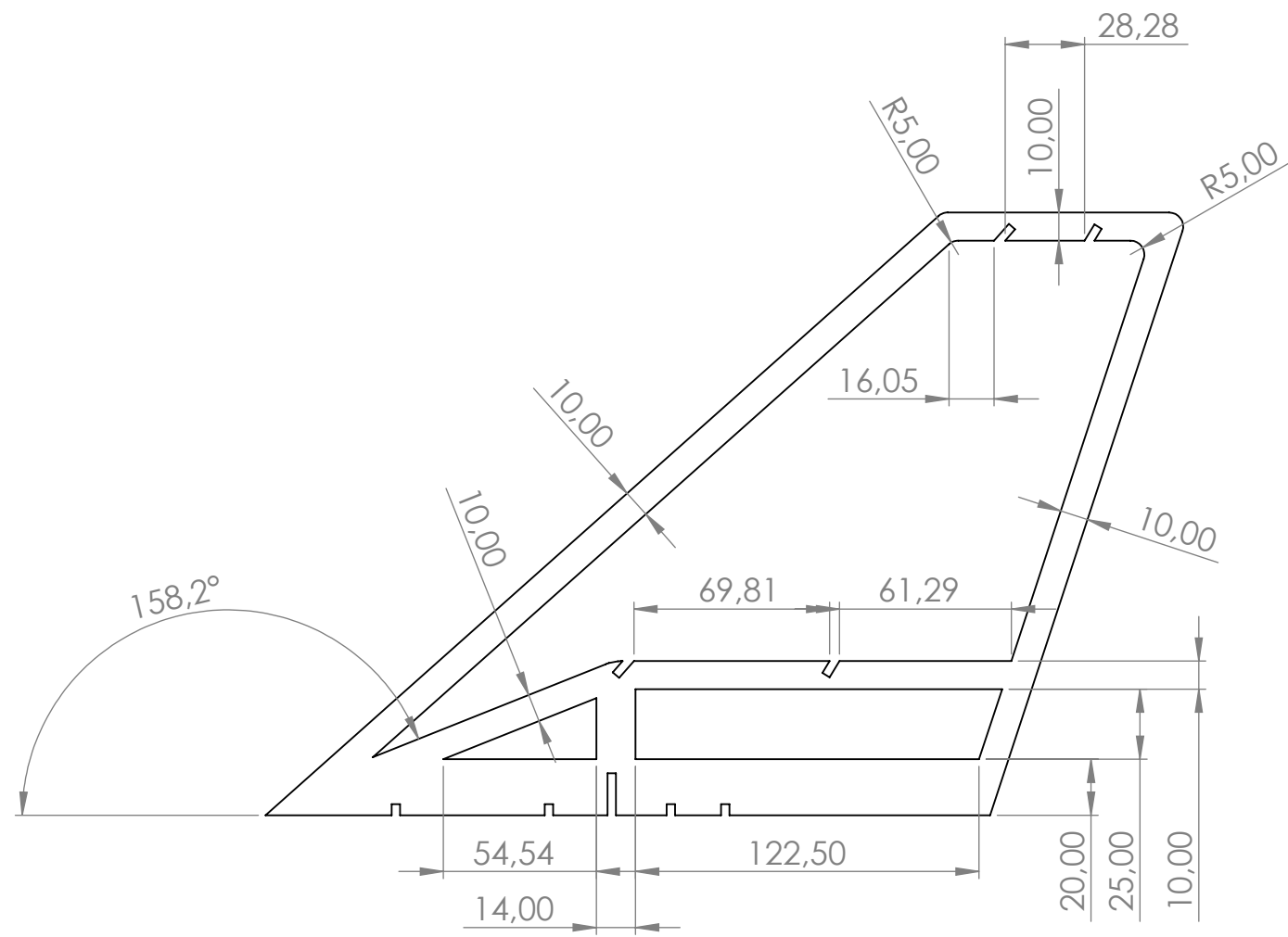
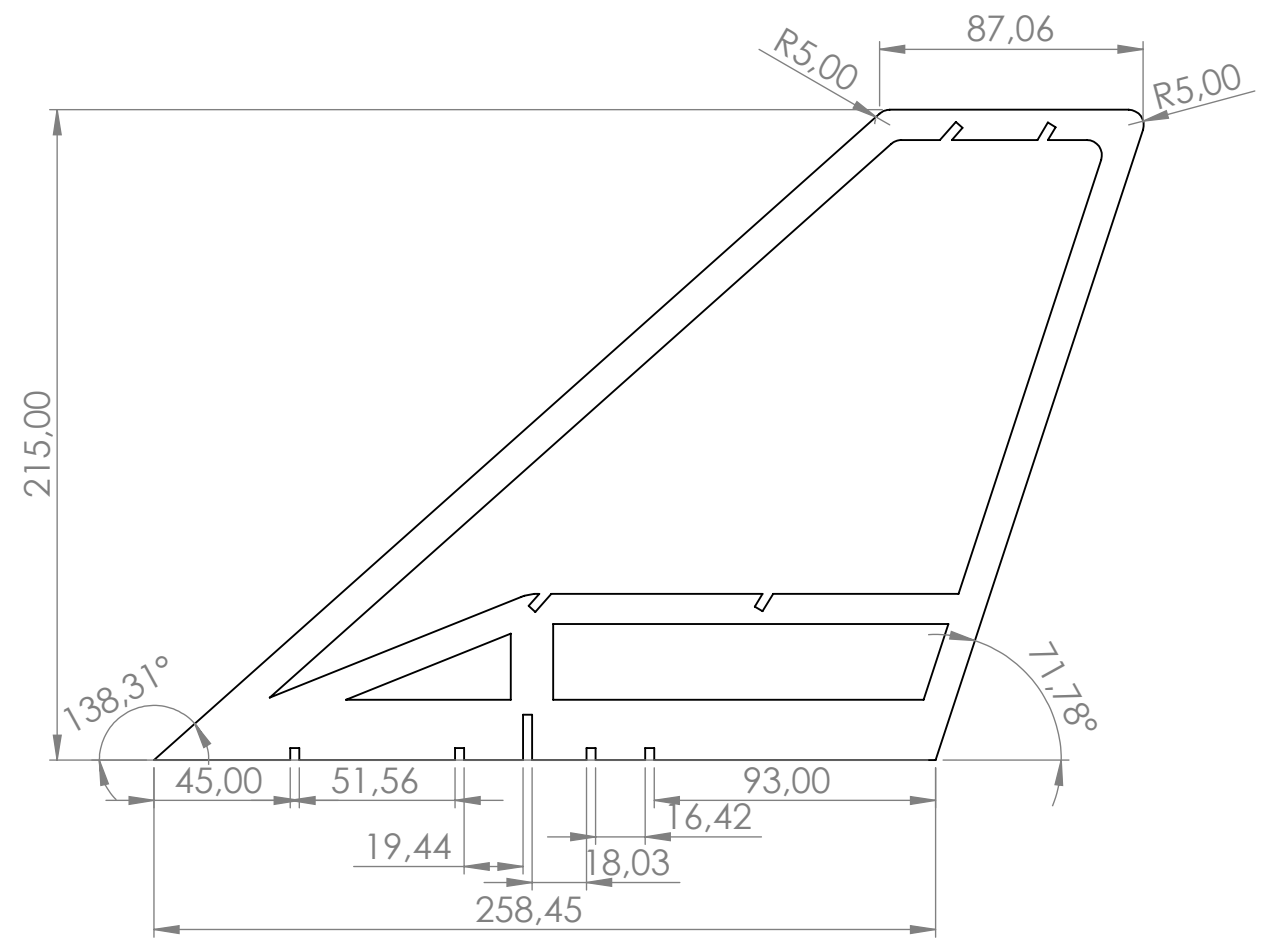
VS-3



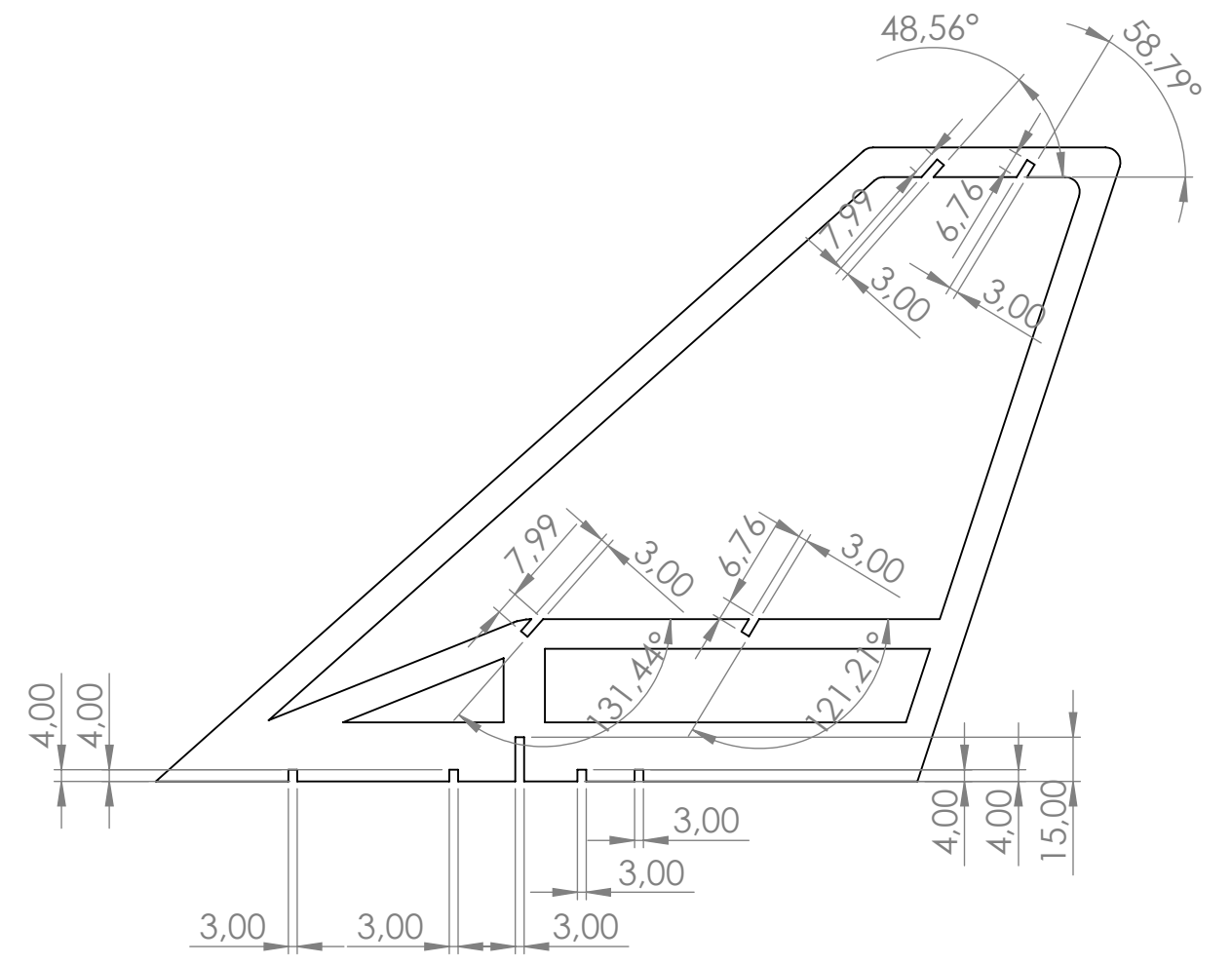
For the vertical stabilizer, two pieces of each will be needed to build the scale model.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:1	Date: 21/06/2023	Size: A3
Sheet: 18	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

VS-4

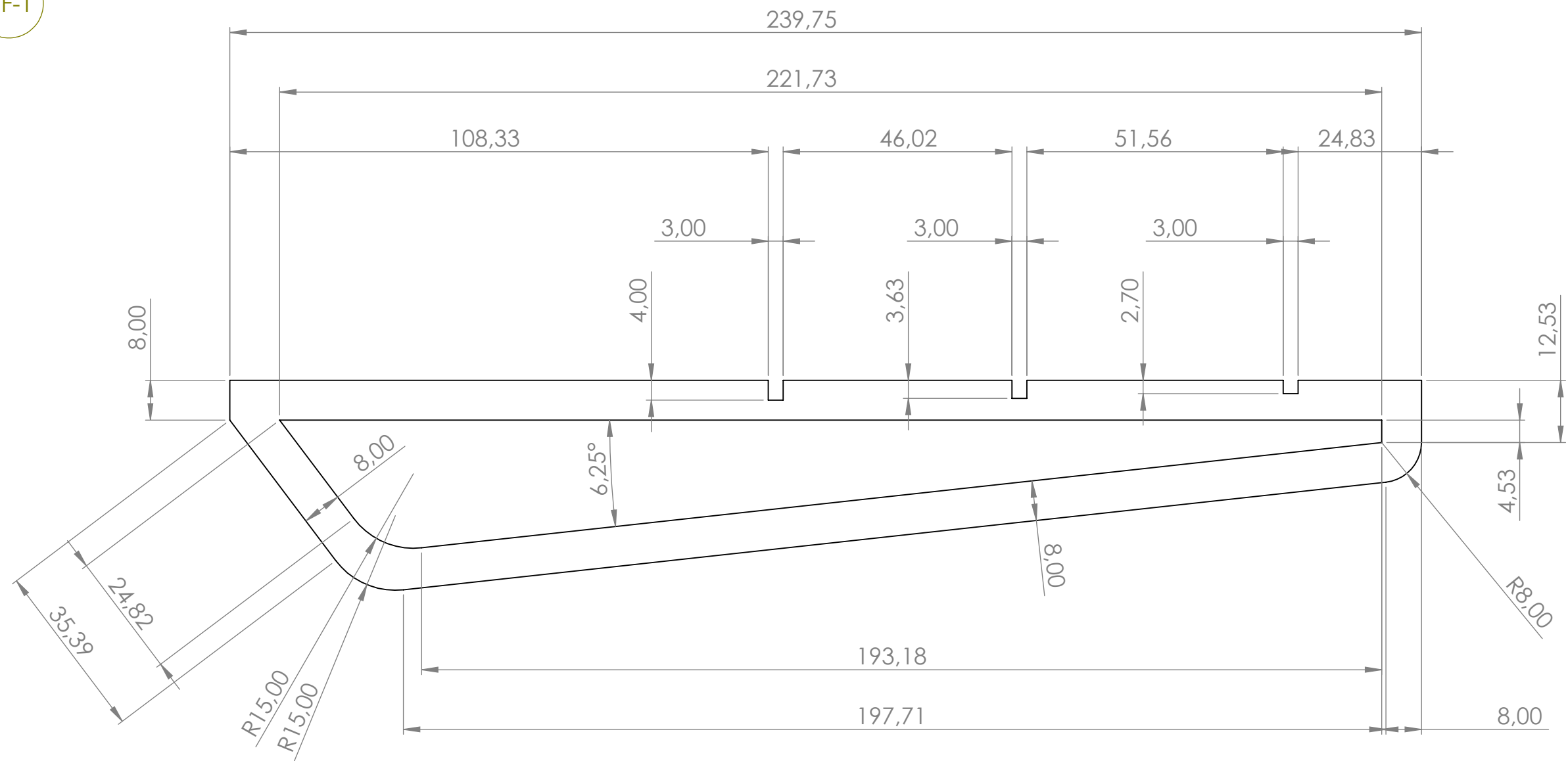


Note that three different drawings have been made of this part in order to incorporate all dimensions clearly. For the vertical stabilizer, two pieces of each will be needed to build the scale model.



TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:2.5	Date: 21/06/2023	Size: A3
Sheet: 19	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

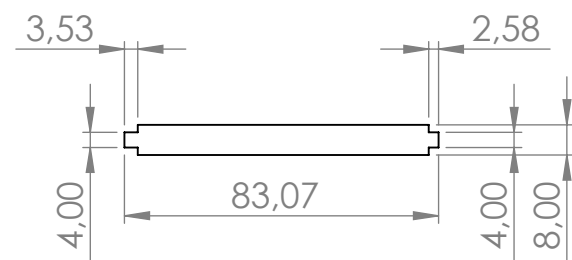
VF-1



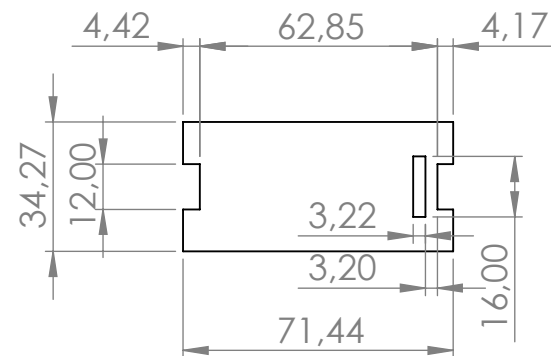
For the ventral fins, two pieces will be needed to build the scale model.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:1	Date: 21/06/2023	Size: A3
Sheet: 20	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

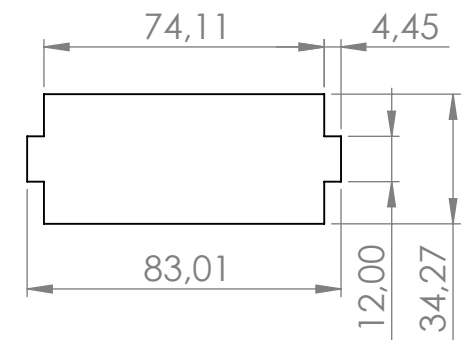
HS-1



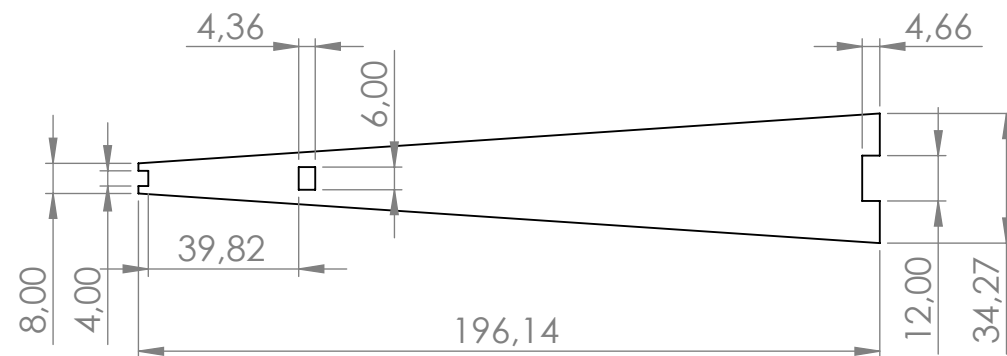
HS-2



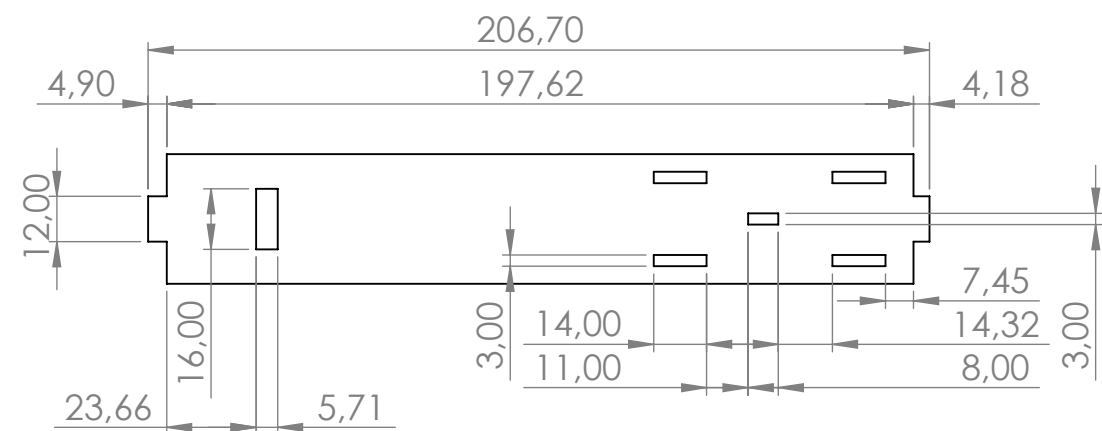
HS-3



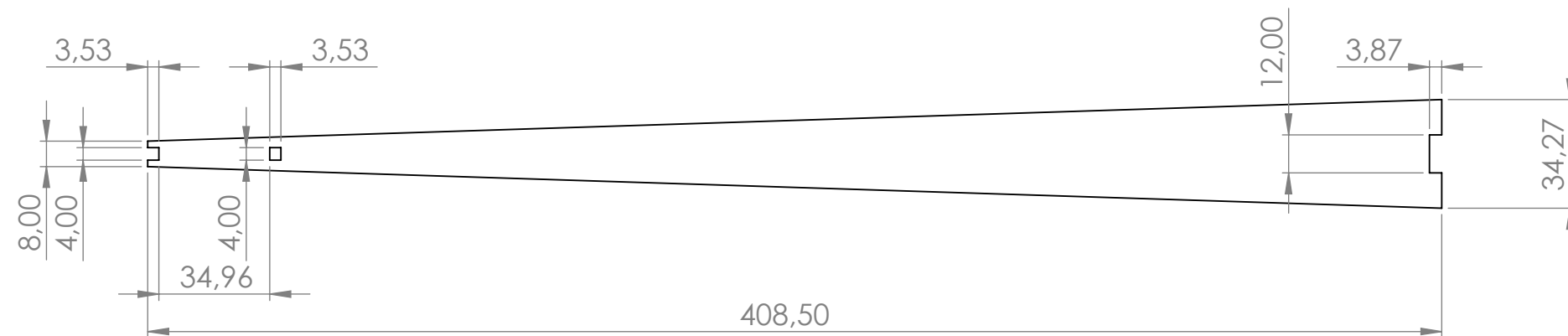
HS-4



HS-5



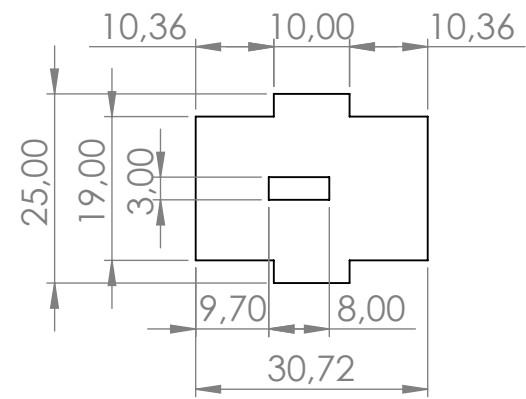
HS-6



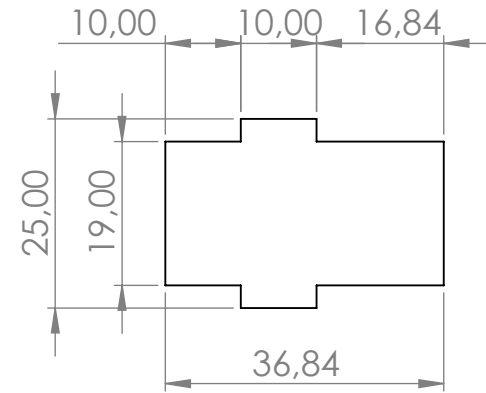
For the horizontal stabilizer, two pieces of each will be needed to build the scale model.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:2	Date: 21/06/2023	Size: A3
Sheet: 21	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

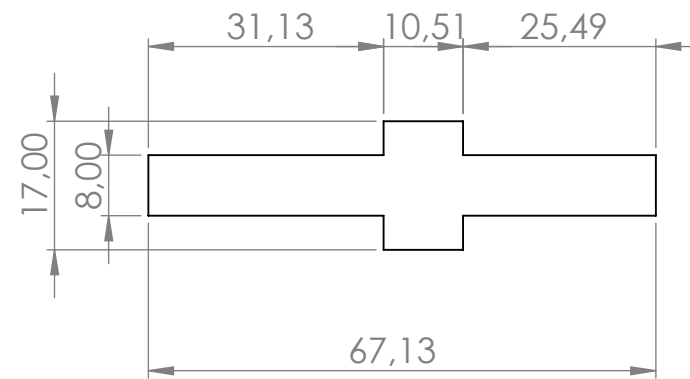
HS-7



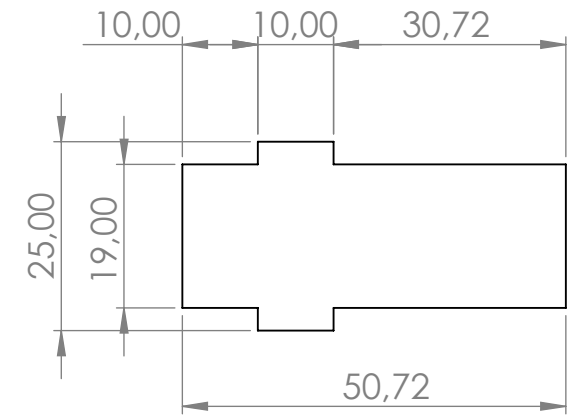
HS-8



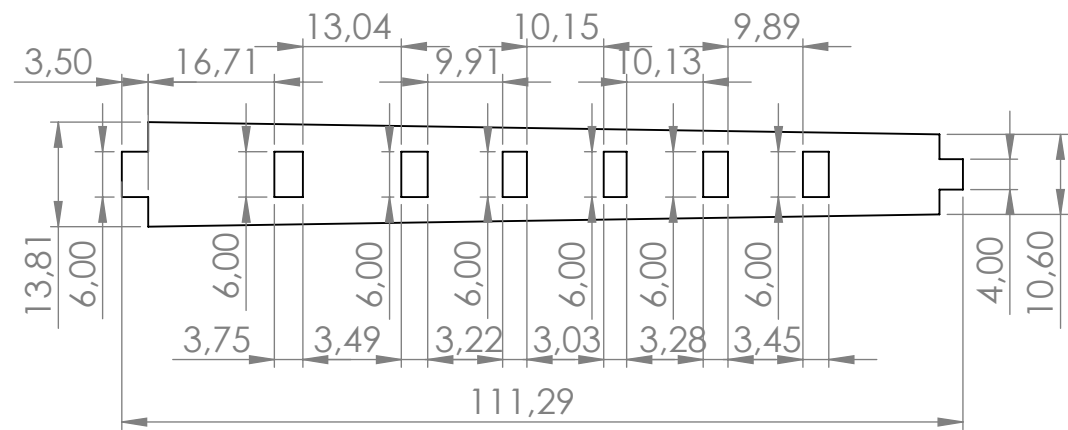
HS-9



HS-10



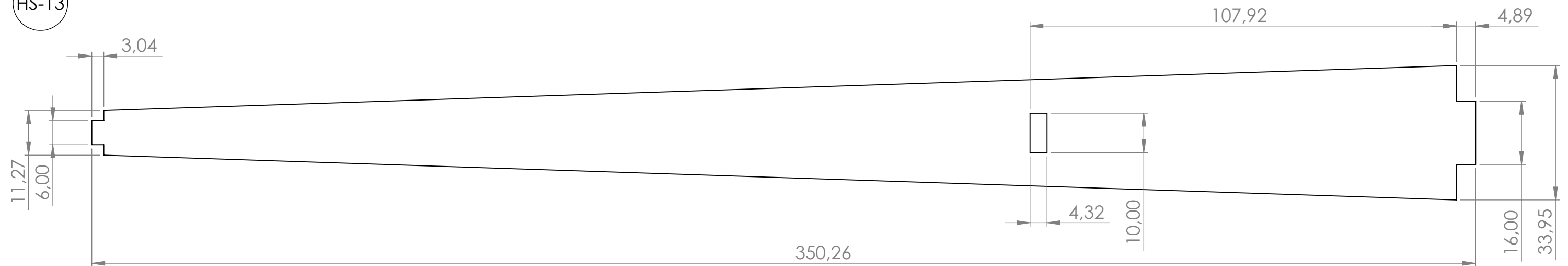
HS-11



HS-12



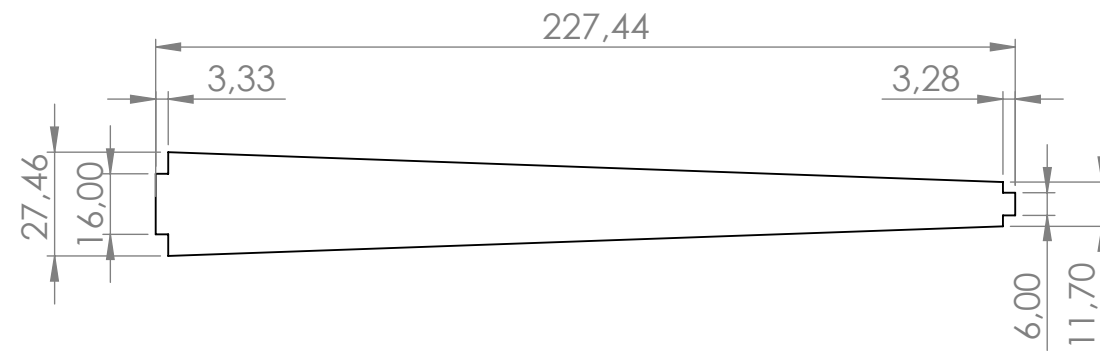
HS-13



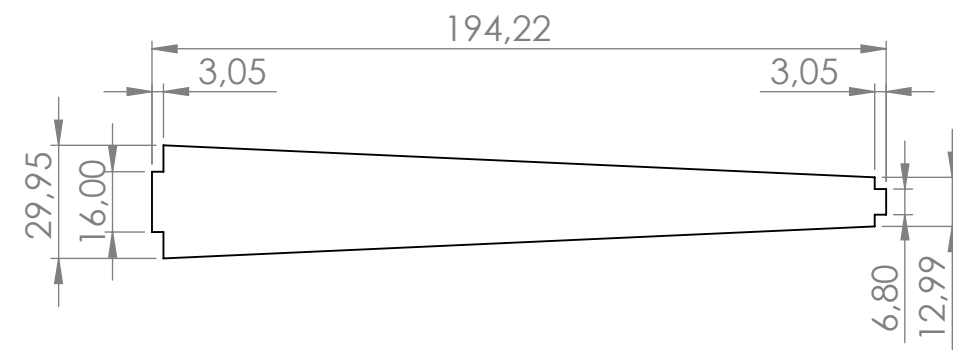
For the horizontal stabilizer, two pieces of each will be needed to build the scale model.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:1	Date: 21/06/2023	Size: A3
Sheet: 22	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

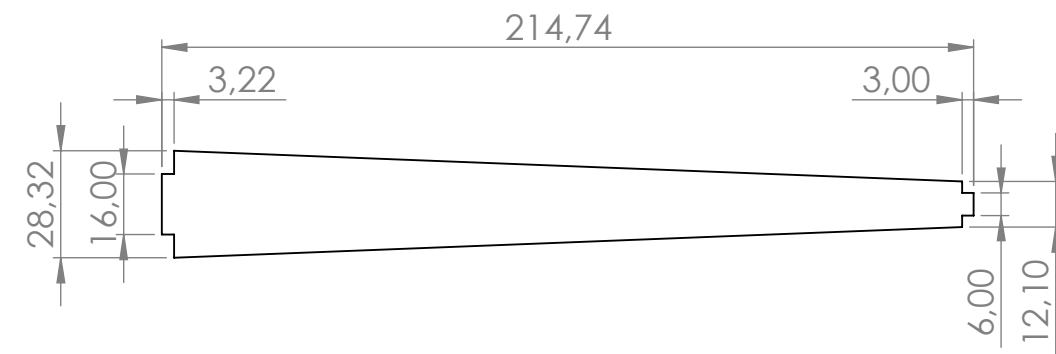
HS-14



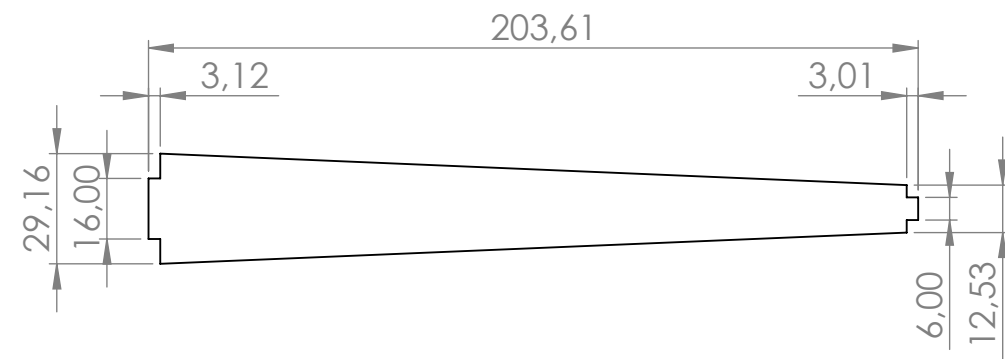
HS-15



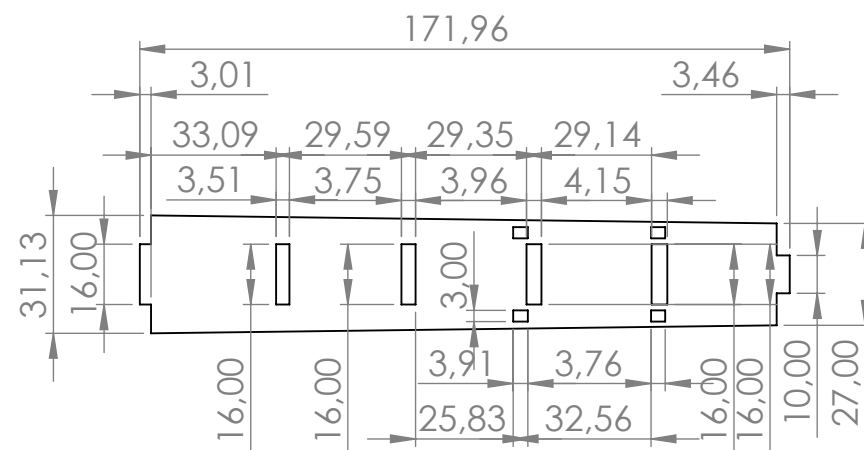
HS-16



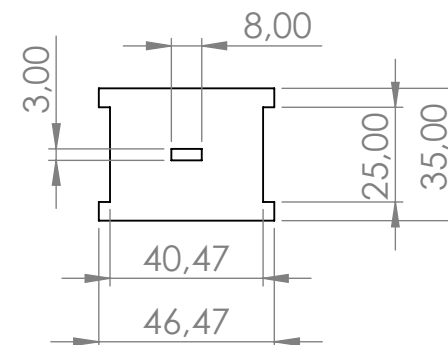
HS-17



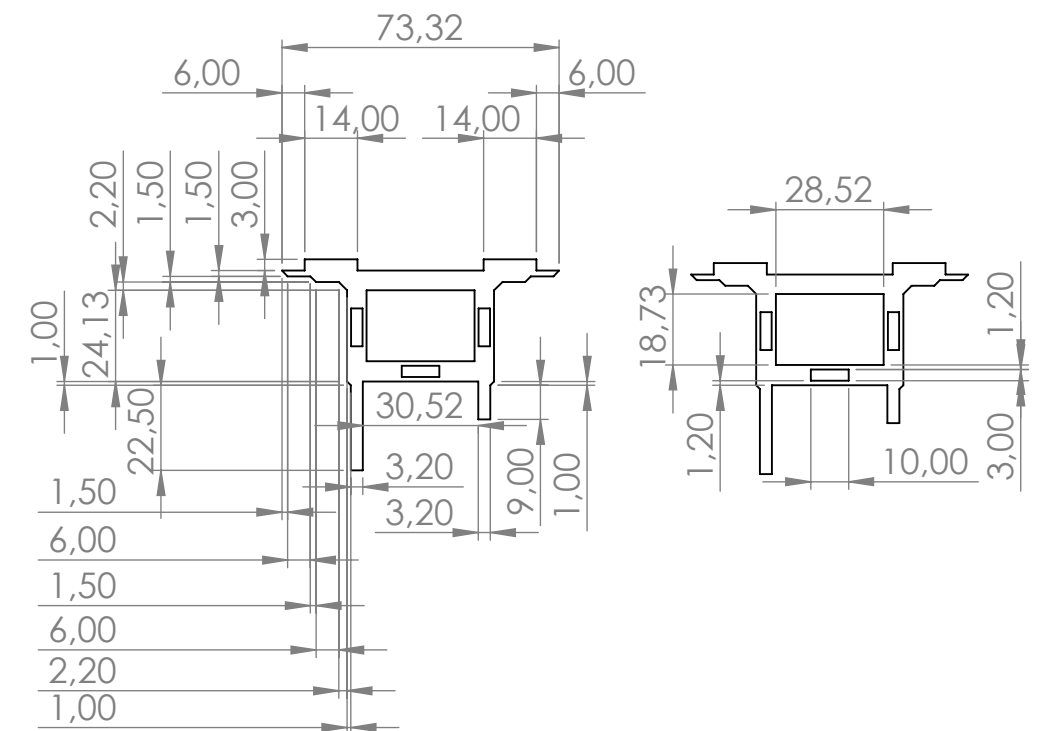
HS-18



HS-19



HS-20



For the horizontal stabilizer, two pieces of each will be needed to build the scale model. Except for HS-19, of which four are needed.

TFG	Study of the F-14 Tomcat structure and reproduction of a 1:12 scale model	
Scale: 1:2	Date: 21/06/2023	Size: A3
Sheet: 23	Name: Sabina Campos i Fuentes	UPC - ESEIAAT

7 AutoCAD drawings

In this section, the images of drawings in .dwg format generated in AutoCAD are attached. This was the format required by the provider. These drawings follow all the pre-cutting requirements imposed by the service provider and are detailed in the written project report.

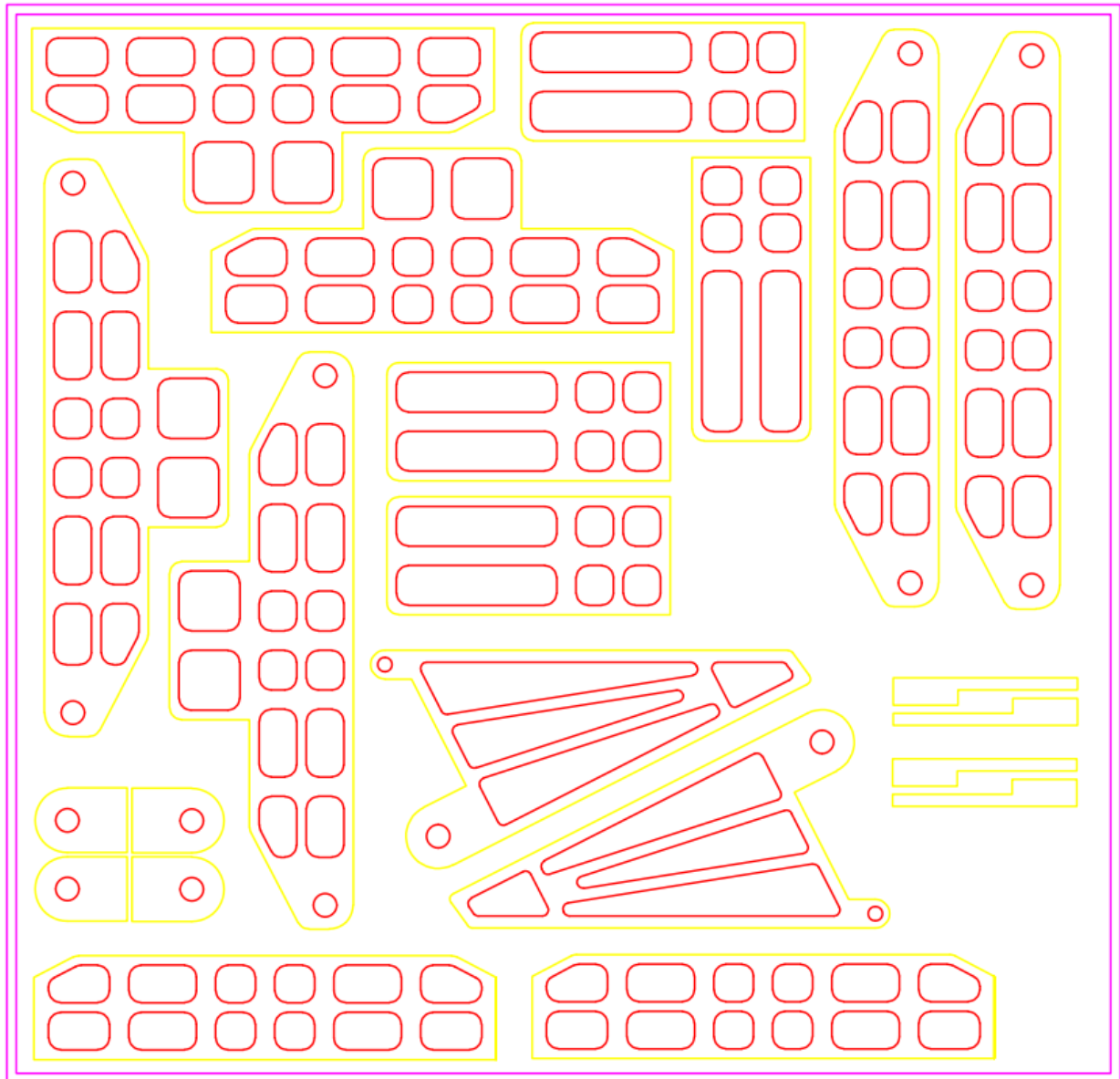


Figure 1: Image of the first AutoCAD drawing with dimensions 930 x 900.

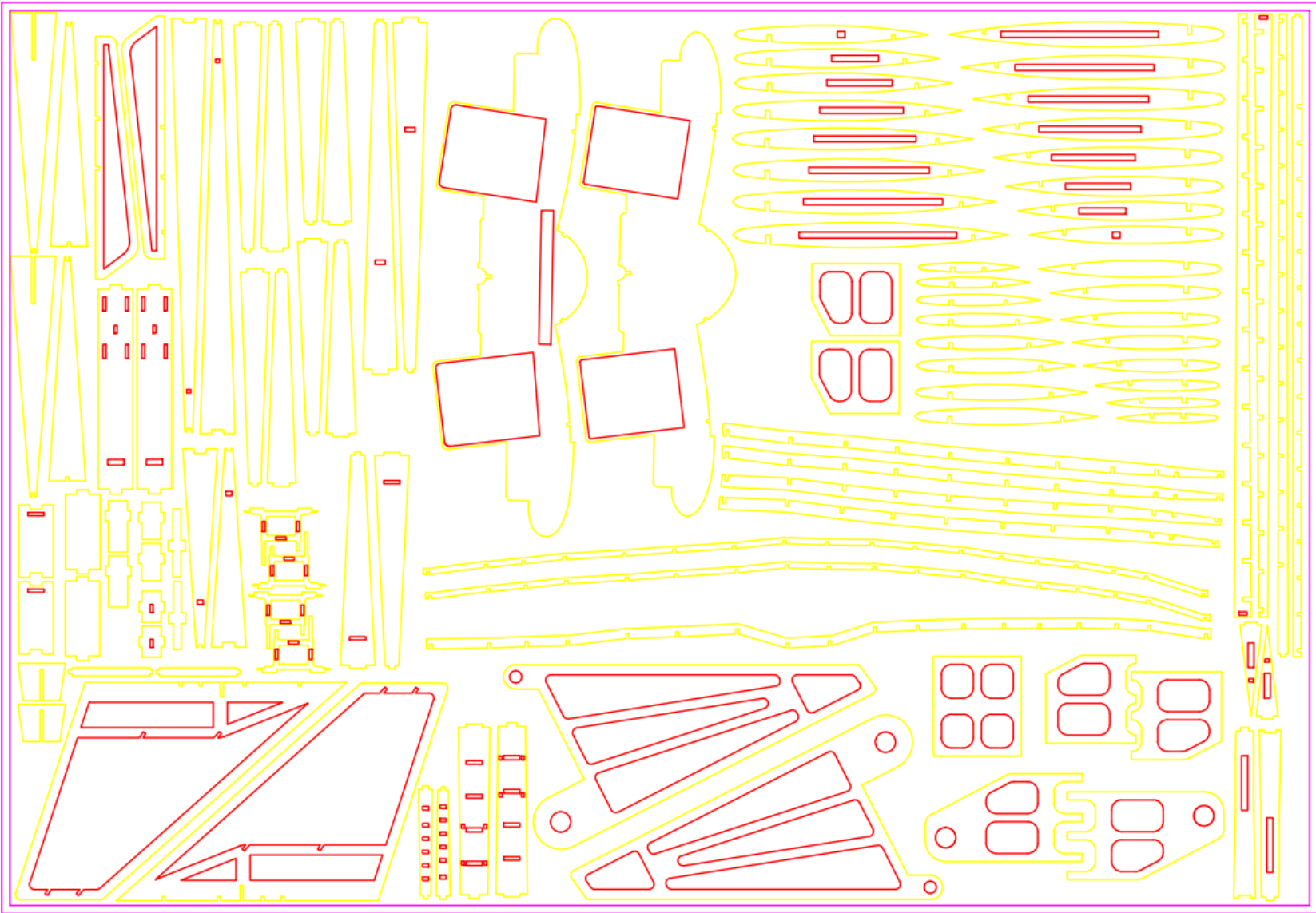


Figure 2: Image of the second AutoCAD drawing with dimensions 1300 mm x 900 mm.

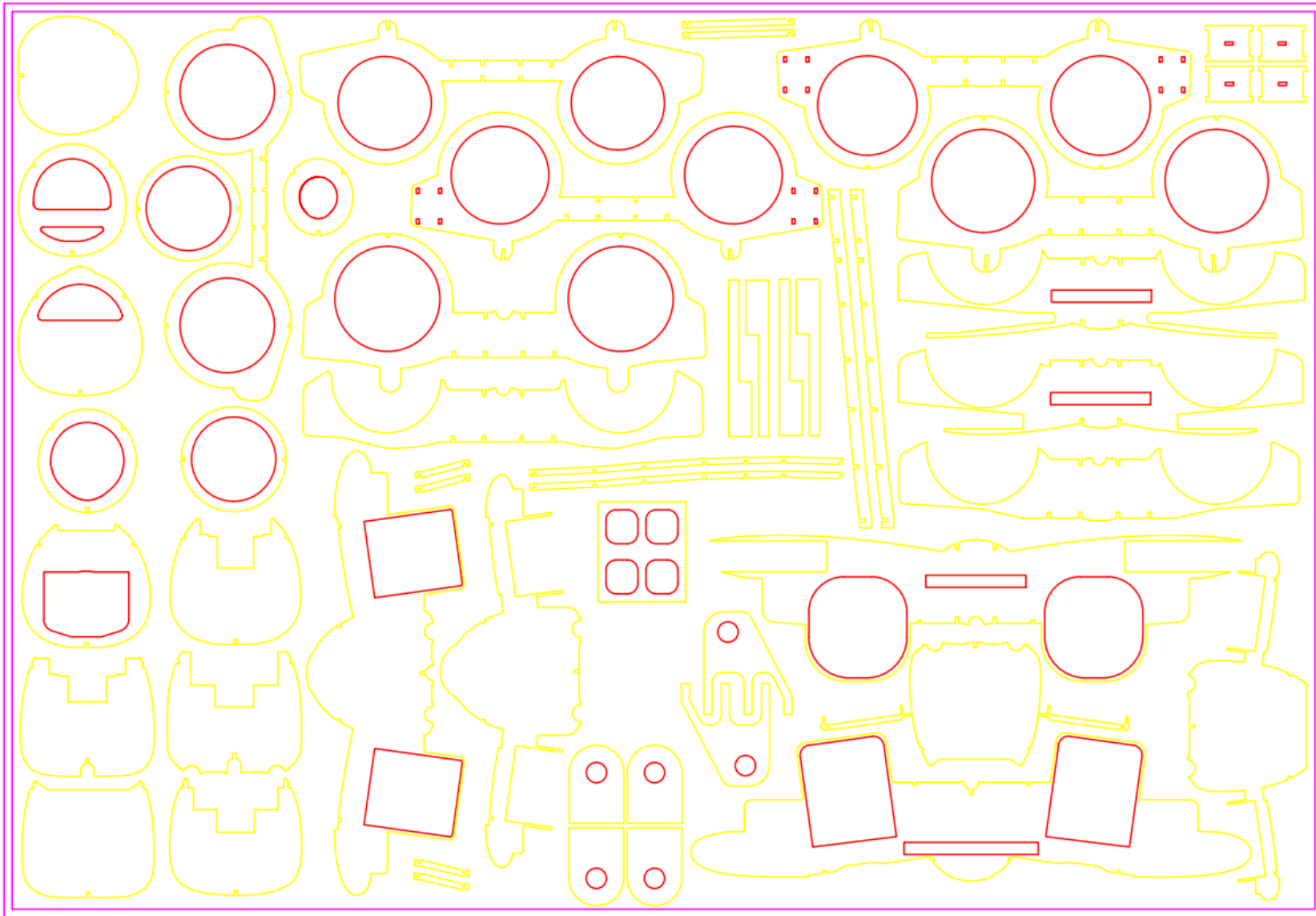


Figure 3: Image of the third AutoCAD drawing with dimensions 1300 mm x 900 mm.